

Government of Assam

**JICA ASSISTED GUWAHATI WATER SUPPLY PROJECT
(GWSP)**

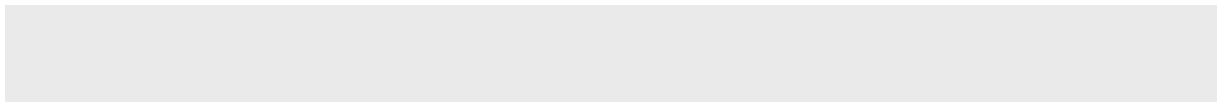
**JICA ODA LOAN PROJECT
(ID-P201)**

**QUALITY ASSURANCE
AND QUALITY
CONTROL MANUAL**

January 2011

Government of Assam

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(GWSP)**



PREFACE

This Quality Assurance/Quality Control Manual has been prepared with the objective of putting in place a comprehensive, consistent and common system for quality assurance and quality control during implementation of the *Guwahati Water Supply Project (JICA ODA Loan No. ID-P-201)*. The Manual covers the overall quality assurance system and the field level quality control procedures for different types of works, based on the technical specifications adopted for the Project.

The QA/QC Manual covers tests and inspections for different materials and works. The acceptance/rejection limits have to be assessed in relation to the technical specifications as incorporated in the construction contracts. *If there are any differences between the tests and other quality control procedures described in the Manual and those stipulated in the Project's contract documents and specifications, the later shall prevail.*

It is suggested that the Manual be kept in a ring binder so that any additions or modifications can be easily incorporated later, if required.

Based on the nature of different contract packages, the relevant portions can be used by contractors, Executive Agencies and PMC for day-to-day reference.

GUWAHATI WATER SUPPLY PROJECT (JICA ODA LOAN NO: ID-P-201)

QUALITY ASSURANCE/QUALITY CONTROL MANUAL

TABLE OF CONTENTS

PREFACE	I
TABLE OF CONTENTS	III
LIST OF ACRONYMS	VII
1 INTRODUCTION	1
1.1 BACKGROUND	1
1.2 QUALITY DEFINITIONS	1
1.3 QA/QC MANUAL	1
2 ORGANIZATION	3
2.1 PROJECT IMPLEMENTATION ARRANGEMENTS	3
2.1.1 General responsibilities	10
2.1.2 Quality assurance/quality control duties	10
3 DESIGN CONTROL	13
3.1 DESIGN PREPARATION PROCESS	13
3.2 DESIGN CHECKING REVIEW PROCESS	13
3.3 DESIGN REVIEW CHECKLIST	14
4 CONSTRUCTION QUALITY CONTROL – GENERAL	15
4.1 INTRODUCTION	15
4.2 TESTING	15
4.3 SITE INSPECTIONS	17
4.3.1 Day-to-day supervision	17
4.3.2 Periodic quality inspections	17
4.3.3 Squad checks	18
4.4 QUALITY CERTIFICATION AND ACCEPTANCE	18
5 CONTROL OF MATERIALS AND EQUIPMENT COMPONENTS	19
5.1 GENERAL	19
5.2 MATERIALS TESTED ON SITE	19
5.3 MATERIALS AND EQUIPMENT CERTIFIED BY MANUFACTURER	25
5.4 MATERIALS AND EQUIPMENT INSPECTED BY THIRD PARTY	27
6 CONTROL OF GENERAL CIVIL AND STRUCTURAL WORKS	29
6.1 CONSTRUCTION SEQUENCE AND CONTROL FLOW CHARTS	29
6.2 TESTING OF WORKS	29
6.3 INSPECTION CHECKLISTS	29
7 CONTROL OF ROAD WORKS	35
7.1 CONSTRUCTION SEQUENCE AND CONTROL FLOW CHARTS	35
7.2 TESTING OF WORKS	35
7.3 INSPECTION CHECKLISTS	35
7.4 GUIDELINES FOR PAVEMENT LAYERS	45

7.5	TOLERANCES	47
7.5.1	Horizontal alignment tolerances	47
7.5.2	Surface levels tolerances	47
7.5.3	Surface regularity of pavement courses	48
8	CONTROL OF PIPELINE WORKS.....	49
8.1	CONSTRUCTION SEQUENCE AND CONTROL FLOW CHARTS	49
8.2	TESTING OF WORKS	49
8.3	INSPECTION CHECKLISTS	49
9	CONTROL OF ELECTROMECHANICAL WORKS	55
10	DOCUMENT CONTROL.....	56
10.1	DESIGN DOCUMENT CONTROL.....	56
10.1.1	Turnkey contracts.....	56
10.1.2	Item rate contracts (Working Drawings)	58
10.2	TEST REPORT CONTROLS.....	58
10.3	TRACKING OF INSTRUCTIONS	59
10.4	SITE ORDER BOOK	59
10.5	CORRESPONDENCE CONTROL.....	59
10.6	INSPECTION OF SITE DOCUMENTATION	60
11	REPORTING.....	61
11.1	TYPES OF PROGRESS REPORTS	61
11.2	REPORTING SCHEDULE AND DISTRIBUTION	61
11.2.1	Contractors' progress reports	61
11.2.2	Monthly contract reports	61
11.2.3	Monthly finance reports	61
11.2.4	Project management consultant's progress reports	61
11.2.5	Quarterly progress reports.....	62
11.3	REPORT FORMATS	62

APPENDICES

Appendix A: Test Report Formats

Appendix B: Inspection Checklists

Appendix C: Documentation Formats

Appendix D: Monthly Report Formats

D.1 Format for Contractors' Monthly Progress Reports

D.2 Format for Engineers' Monthly Reports

D.3 Format for PIU Monthly Financial Report.....

D.4 Format for Design and Supervision Consultants' Monthly Progress Reports

1. INTRODUCTION

This section of the Quality Assurance/Quality Control Manual presents the Project's background, defines quality-related terms, and gives an outline of the Manual.

1.1 Background

Guwahati Metropolitan Development Authority has undertaken the implementation of Guwahati Water Supply Project (JICA Loan IN-P201).

Infrastructure facilities to be provided include Intake structure, rising main, water treatment plants, reservoirs supply mains and distribution systems. The Project Implementation Unit have been setup. The Project Management Consultant have been appointed and have started functioning from the month of February 2010.

1.2 Quality Definitions

Quality is the totality of features and characteristics of a product or service that bear on its ability to satisfy a given need. The term 'given need', in case of project works, can be interpreted as the functional requirements. The quality of outputs is always agreed upon between the supplier and the client (in project works, contractor and the owner), and the quality objective must be to achieve zero defects within acceptable tolerances. It can be made possible only by ensuring the quality at all stages of project works.

The following are some definitions pertaining to quality and how to achieve it:

- Quality: Conformance to requirements.
- Quality Control (QC): The operational techniques and activities (such as reviewing, checking, inspecting, testing, etc.) that are used to fulfill requirements for quality.
- Quality Assurance (QA): The planned and systematic actions necessary to provide adequate confidence that the work will satisfy quality requirements.
- Quality System (QS): A set of documented processes, which seek to provide confidence that the project outputs will fulfill the functional requirements. The Quality System should encompass the organization, responsibilities, human resources, materials, equipment, processes, inspections, testing and other parameters of the project. A key element of QS is the QA/QC Manual.
- Quality Surveillance: This normally covers two aspects:
 - At the project level, a review to ensure that the quality practices are implemented and documented in relation to the quality system; and
 - At the contract package level, inspection and testing to ensure that the works executed meet the required quality standards.

1.3 QA/QC Manual

This QA/QC Manual focuses on the implementation activities of the project following contract award, and primarily on supervision and quality control of construction works. Other aspects of project implementation are also covered but in less detail. The QA/QC Manual is intended to be used primarily by the project staff of the PMC and the Contractors.

The QA/QC Manual for the Project does not attempt to suggest technical specifications, since these are stated in the contract documents. Its aim is to ensure that the works are executed as per specifications, i.e. it is looked at as a means to achieve the end results. Quality control and test results shall be interpreted as applicable for different packages, in accordance with the contract conditions.

The subsequent sections of this Manual are as follows:

- Organization
- Design Control
- Construction Quality Control – General
- Control of Materials and Equipment Components
- Control of General Civil and Structural Works
- Control of Canal Works
- Control of Road and bridge Works
- Control of Pipeline Works
- Control of Electro-mechanical Works
- Document Control
- Reporting

2. ORGANIZATION, RESPONSIBILITIES AND AUTHORITIES

This section of the QA/QC Manual describes the organizational arrangements for project implementation and outlines the responsibilities of each organization.

2.1 Project Implementation Arrangements

- a) As agreed between the GMDA and JICA, the Water Supply Facilities will be executed through the Project Implementation Unit (PIU) of GMDA. The State has appointed a Project Director for the Project. PIU has the overall responsibility for coordination and management of the Project activities, including Project design, implementation, budgeting and financial planning, benefit monitoring and evaluation, socioeconomic surveys, environmental assessment and protection, institutional and policy development, community participation and coordinating the work of all consulting services under the Project.
- b) Project Implementation Units (PIU) have been established to support the project works. They have the primary responsibility for planning and implementing all the Project components in time, within cost estimates, and to the quality standards specified in the contract documents within their respective areas.
- c) The GMDA has recruited an International Consulting Agency as Project Management Consultant (PMC). He is responsible for assisting the PIU in implementing, managing and monitoring Project activities, recommend ways to accelerate Project implementation.
- d) The PMC is responsible for all Project Management activities in the respective cities along with the PIU. This includes identification of the work packages, design, tender document preparation, evaluation and implementation. They are also responsible for quality assurance.
- e) The details of the individual responsibilities of each wing related to various activities are laid down as below to avoid any misunderstanding. The list of responsibilities as enumerated below may all not appear to be of direct bearing on the quality, but they do contribute to clarity of responsibility and hence help organize the works better. This is only to facilitate smooth functioning and is not meant in anyway to dilute the responsibilities and authority various wings have in their respective jurisdiction under the contract documents.

Table 2.1: Responsibilities of Key Agencies

Sl. No.	TASK	RESPONSIBILITY	Unit
1.0	Design of works and Contract Packaging		
1.1	Basic Design of works for Turnkey Contracts with drawings and estimates	Preparation	PMC
		Review approval	PIU
1.2	Detailed designs and drawing of turnkey contracts	Preparation	Contractor
		Check, review & recommend	PMC
		Approval	PMC/PIU

1.3	Detailed design, drawings and estimates of Item-Rate contracts	Preparation Review & approval	PMC PIU
1.4	Contract packaging of works and goods	Review & recommend Approval	PMC/PIU JICA
1.5	Modification of scope of works	Recommend Approval	PMC/PIU JICA
2.0	TENDER DOCUMENTS		
2.1	Documents for ICBs/LCBs	Preparation, Review and recommend Approval	PMC PMC/PIU PIU/JICA
3.0	TENDER AND AWARD		
3.1	Prequalification of contractors	Notice inviting PQ applications Receiving applications Evaluation and Short listing Review & recommend Approval	PIU/GMDA PIU/GMDA PMC/ PIU Evaluation committee PIU/GMDA JICA
3.2	Tenders for works & goods with estimated value more than Rs. 1.0 Crore and less than US \$ 10.0 million (LCB packages)	NIT/receiving tenders Comparative statement Bid examination Evaluation Review & recommend Approval	PIU PMC PMC/PIU PMC/ PIU Evaluation committee PIU/GMDA JICA

3.3.	Issue of Work Orders, Signing of Agreements and upkeep of original contract documents for all works and local procurement.		PIU/GMDA
3.4	Issue of Purchase orders, signing of agreements and upkeep of original contract documents for centralized procurement.		PIU/GMDA
4.0	CONTRACT ADMINISTRATION AND CONSTRUCTION SUPERVISION GENERAL	PRIMARY SECONDARY	PMC/PIU
4.1	overall administration and management of the contracts including interpretation of the technical specifications and other contract documents as may be required;	Primary Secondary	PMC PIU
4.2	Review and revise construction drawings as may be necessary from time to time, either to suit site conditions, changes in construction strategy or changes in design, and provide clarifications / explanations on the designs and drawings to the contractor.	Submission Review, check and recommend Approval	PMC/ Contractor PMC PIU
4.3	Review and recommend acceptance or modification of the construction drawings and designs prepared by the contractors for the turnkey contracts	Review, check and recommend Approval	PMC PIU
4.4	Giving level and layout for those items of work where dimensional accuracy has a direct bearing on the quality and performance of the finished work to ensure conformity with the quality requirements stipulated in the contract;	Primary Secondary	PMC PIU
4.5	Assess the adequacy of the inputs such as materials, labor and equipment provided by the contractor and the construction methods proposed and ensure that they are satisfactory with reference to the technical requirements, implementation schedule, environmental aspects, and safety of the works, project personnel, and general public welfare should be only in case a dispute arises.	Primary Secondary	PMC PIU
4.6	Proper inventories and accounts are maintained of all dismantled materials, particularly for those materials which are to be re-used in the works;	Primary Secondary	PMC PIU
4.7	The work site is maintained in a neat, orderly and safe manner;	Primary Secondary	PMC PIU
4.8	Any inconvenience to the public is minimized; and Payments are made to the Contractor in a timely manner.	Primary Secondary	PIU PMC
4.9	Necessary assistance to solve any contractual dispute and sort out issues requiring external interdepartmental coordination, which has an overall obligation to ensure the successful implementation of the project, works.	Primary Secondary	PIU PMC
4.10	Providing continuous on-site supervision during	Primary	PMC

	construction and ensuring the safety of the works;	Secondary	PIU
4.11	Supervising and monitoring the progress of the works, including identifying cause(s) of delays, determining remedial actions to correct such delays, and issuing instructions to contractors;	Primary Secondary	PMC PIU
4.12	The contractor fulfills his obligations under the contract and satisfactorily completes all contractual obligations and complies with all applicable statutes, regulations, contract conditions, specifications and instructions;	Primary Secondary	PMC PIU
4.13	The contractor completes the work within the scheduled time	Primary Secondary	PMC PIU
4.14	Ensuring that Site Order Books, Daily Work Records, Labor, Material and Machinery Logs are properly maintained	Primary Secondary	PMC PIU
4.15	Making test records and results available to the PMC/PIU for review and assessment in case of dispute only.	Primary Secondary	PMC PIU
4.16	Acting on project issues and problems as they arise, and promptly issuing written instructions to the contractors to address the problems;	Primary Secondary	PMC PIU
4.17	Ensuring that the contractor properly prepares the "As Built" drawings for the completed works;	Primary Secondary	PMC PIU
4.18	Ensuring that the contractor prepares and submit Monthly Progress Report in the approved format and on time	Primary Secondary	PMC PIU
5.0	QUALITY ASSURANCE AND INSPECTIONS	Primary Secondary	PMC PIU
5.1	Prepare a simplified Quality Control Manual for use of the field staff, and assist in providing on-the-job training to PIU and Contractor staff	Primary Secondary	PMC PMC
5.2	Provide effective supervision of the works in order to ensure the quality and conformity with the standards and specifications prescribed in the contract	Primary Secondary	PMC PIU
5.3	Regular and frequent inspections of all work sites should be made to check the nature and quality of work being done, to verify the materials, equipment and labor engaged at the site, to review the quality control tests and test results, and to ensure that the work is being implemented in accordance with the approved standards and that the quality control procedures set forth under the contract are being followed.	Primary Secondary	PMC PIU
5.4	Witness all quality control sampling and testing done by the contractor. Compile and review all quality control data obtained from tests conducted by the	Primary Secondary	PMC PIU

	contractor or by others and verify the accuracy of the test data by checking the procedures used in the field for sampling and testing of the materials and works. Instruct & Supervise independent sampling and testing wherever considered necessary, or as may otherwise be required to check and verify the accuracy of the test results conducted by the contractor. Assess the test results and recommend on acceptance of the materials supplied and on the works completed. To ensure that proper records of the tests conducted are maintained.		
5.5	Periodic inspections on an as-required basis to inspect and accept interim work completion stages (i.e., completion of excavator in order to permit the contractor to proceed with further works. All approvals should be entered into the site order book and signed by all parties, and no work on further stages should be permitted until the earlier stage work has been inspected and accepted. In the event that the work fails to meet the required standards, any removal and replacement or other remedial measures which may be required should be clearly explained along with a time schedule for completing such work.	Primary Secondary	PMC PIU
5.6	Joint inspection(s) of the completed works by PMC, PIU and contractor, preparing a statement of exceptions for any works which may remain to be completed, approving and accepting the completed works, issuing the Certificate of Acceptance and making final payment to the contractors..	Primary Secondary	PMC PIU
5.7	Inspections of the completed works, ensuring that any defects in materials or workmanship are properly identified in a timely manner.	Primary Secondary	PMC PIU
5.8	Participate in monthly inspections and site coordination meetings of PIU, PMC and Contractor for all works to review the overall progress and quality of the works, review the problems which may have arisen, the instructions which were issued to the contractor to address these problems and the contractor's compliance with these instructions, and to agree on any further actions which may be required to be taken to improve either the progress or quality of the works. The PMC shall be responsible to prepare the minutes of the site coordination meetings in order to maintain a permanent record of all agreements reached, instructions issued and actions to be taken.	Primary Secondary	PMC PIU
5.9	Following expiration of the Maintenance Period/defect liability period, inspecting the works, identifying any defects in materials or workmanship, issuing the Completion Certificate and releasing the security deposit or balance of security deposit following satisfactory correction of all defects.	Primary Secondary	PMC PIU

6.0	CONTRACT VARIATIONS	Proposal	PMC
		Approval	PIU
6.1	Make a monthly assessment of the progress and quality of the works and recommend to the PIU on any necessary variations to the contracts, including work programs, work procedures, inputs, safety, quality, variation orders, completion dates, and/or any other matters which may affect the timely and satisfactory completion of the work. Propose and present for approval any changes in the plans which may be deemed necessary, and indicate any effect such changes may have on the contract. Assist the PIU in preparing any required variation orders and obtaining necessary approvals from the PIU prior to issuance to the contractor where required.	Proposal Approval	PMC PIU
6.2	Examining all proposed variation orders or claims from the contractor for time extensions, extra compensation, or expenses or other similar matters, preparing variation orders and obtaining necessary approvals from PIU prior to issuance to the contractor where required.	Proposal Approval	PMC PIU
7.0	MEASUREMENT AND PREPARING BILLS AND PAYMENTS	Preparing Approval and Payment	Contractor PMC/PIU
7.1	Conducting with contractor joint measurement of the works in the stipulated format for payment	Preparation /measurement of bills	PMC & PIU
7.2	Record Measurement for such measurements which cannot be verified subsequently	Measurement and record	PMC
7.3	Preparing necessary payment release order of security and payment thereof after completion of the defect liability period as per the contract.	Verification of bills & recommendation for payment	PMC
7.4	Passing of the bills for supplies received under the material procurement packages after due inspection and checking and payment thereof. The PMC, and PIU will provide a certificate regarding full confirmation of the goods to the specifications and being in good condition.	Verification of bills & recommendation for payment Review & Approval	PMC PIU
8.0	REPORTING		
8.1	Monthly Progress report to PMC	Submission	Contractor
8.2	Preparing Monthly Progress Reports in the approved format, including physical and financial progress, problems encountered and actions taken and submitting to the PIU in a timely manner incorporating the contractor's report.	Submission	PMC
8.3	Submit a quarterly Progress report for submission to JICA through PIU		PMC

8.4	Reporting to PIU from time to time regarding overall physical and financial progress of work, with specific mention of problems encountered and actions taken or remedial measures recommended, variation orders approved, anticipated slippage in any item of work, rectification measures recommended, and any specific assistance required from PIU	Submission Review/Comments /approval	PMC PIU
8.5	Submission of the contract completion report summarizing the construction activities and indicating, among other things, contract changes, claims or disputes, or any other substantive matters having an effect on the cost and progress of the works and accurate and complete "As Built" drawings (to be submitted by the contractor) for the completed works	Submission Review/Comments /approval	PMC PIU
8.6	Assist the PIU in preparing the Project Completion Report	Preparation Review & finalise	PMC PIU
9.0	BUDGET		
9.1	Annual plan of work and forecast of funds requirement for each component of the Project.	Submission Review	PMC PIU
9.2	Annual Project plan and forecast of funds requirement for the Project	Submission Review	PMC PIU
10.0	FINANCE AND MAINTAINANCE OF ACCOUNTS		
10.1	Financial Management Control and maintenance of Project Accounts	Primary Secondary	PIU PMC
10.2	Preparation and submission of disbursement claims	Primary Secondary	PIU PMC
10.3	Compilation, disbursement preparation and submission of reimbursement claims for the Project to the JICA	Primary Secondary	PIU PMC
11.0	INTER DEPARTMENTAL AFFAIRS		
11.1	Land acquisition	Identification & Acquisition	PIU/Land Deptt.
11.2	Identification of Power Connection, Railway crossing, and Road crossings. Pipe/Sewer interconnections with existing systems, permission for use of Forestland, etc.	Primary Secondary	PMC PIU/Line Agency
11.3	Obtaining permissions from other departments and organizing the works as required through them.	Primary Secondary	PIU/Line Agency PMC
12.0	TRAINING		
12.1	Review and assess training needs / requirements of the PIU.	Primary	PMC

		Secondary	PIU
12.2	Development of a comprehensive training plan to meet identified needs.	Primary	PMC
		Secondary	PIU
12.3	Preparation of Training manuals and modules.	Primary	PMC
		Secondary	PIU
12.4	Providing on the job training and conducting overseas training for PIU & PIU Staff	Primary	PMC
		Secondary	PMC

Note: The above procedures are prepared with a view of ensuring a smooth action in various activities, which generally have an overlapping responsibility. They are only a clarification on the responsibilities as prescribed in the respective contract documents. In case of variance, the contract documents will precede over the above stipulations. **Wherever the responsibilities are shown to more than one agency, the order of precedence of responsibility is from left and that of authority is from right.**

Contractors are responsible for the execution of the works in conformance with the requirements of the contract documents.

2.1.1 General responsibilities

Contractors are responsible for providing:

- all necessary plants, labors, equipments and construction materials to be used in the works;
- all plants, equipments, materials and labors for temporary and auxiliary works;
- all equipments and components to be installed or incorporated in the works;
- transportation and storage facilities for all materials and equipment.
- office and accommodation for staff and labors; and for consultants and employer's staff
- sanitation and other basic facilities at the site; and
- all necessary staff and equipment for testing and quality control including site office laboratories.

Contractors are responsible for executing and completing the works in accordance with the specified standards and specifications, within the contractual time allowed, and within the contract price for these works. On water supply turnkey contracts, contractors are also responsible for preparing detail design documents, drawings and obtaining their approval.

2.1.2 Quality assurance/quality control duties

The contractor's QA/QC duties are summarized in Table 2.1. Other duties shall be performed as stipulated in the contract documents or directed by the Engineer/PIU.

Table 2.1

List of Contractor's QA/QC Duties

Activity/Item	Contractor's QA/QC Duties
Designs for turnkey contracts	• Prepare designs using appropriate QA/QC procedures • Submit designs and drawings to the PMC for review and approval • Maintain design document and drawings register at site • Use only approved drawings for construction
Designs for item-rate contracts	• Maintain drawing register at site • Use only approved drawings for construction
Test laboratory and equipment	• Intimate PMC the details, date of completion with requisite manufacturers' and calibration certificates • Maintain the equipment in good condition and calibrate as necessary
Material receipts	• Enter receipts in material register • Intimate PMC in writing
Materials testing	• Prepare mix designs as required by contract and submit test results to PMC • Take test samples in presence of PMC/PIU when requested • Perform materials tests • Submit test reports to PMC with monthly reports • Maintain test log
Rejected materials	• Enter in material register at site • Intimate PMC in writing the proposed date of removal from site and confirm after removal
Material consumption	• Enter daily consumption of materials in material register and indicate balance quantity
Construction equipment	• Intimate PMC the details, date of mobilization along with requisite insurance certificate • Maintain equipment in good working condition
Construction	• Intimate PMC in writing when construction is going to commence and what activities are proposed to be undertaken. • Intimate PMC in advance when critical works, such as concreting, embankment, paving, pipeline laying and jointing, testing, etc., would be undertaken, along with the test

Activity/Item	Contractor's QA/QC Duties
	certificates of the materials proposed to be used in these works. No critical activity shall start unless the material test certificates are verified and approved by PMC. • Provide necessary QA/QC
Daily work progress	• Maintain in daily log
Testing of works in progress	• Perform tests as per contract requirements • Submit test reports to PIU and PMC • Maintain test log
Rejected work items	• Intimate PMC in writing the proposed date of removal from site and confirm after removal, • Rectify defective work and invite PIU and PMC for re-inspection.
Instructions from Engineer	• Enter change orders, site instructions, letters and minutes of meetings issued by the PMC in the Instruction Log
Inspection of Engineer	• Take instructions in Site Order Book. • Advise PIU and PMC of compliance
Progress scheduling and control	• Prepare and maintain project schedules and undertake work in accordance with approved schedule
Reporting	• Prepare and submit Monthly Progress Reports
Records	Maintain the following records on site: • Material Register • Site Order Book • Hindrance/Encumbrances Register • Daily Log • Test Log • Instruction Log • Equipment Register • Labor Register • Approved Construction Drawings • Test Reports • Site Laboratory Record • Permissions Issued by Departments • Correspondence Record • Copies of Monthly Progress Reports • Any other records as specified in the Contract and/or as instructed by the Engineer

3. DESIGN CONTROL

This section outlines the detail design preparation and review processes under the Project and presents basic design review criteria. As discussed in Section 2, the detail design and drawing preparation responsibilities under the Project vary depending on the contracting procedure, as follows:

- For turnkey contracts, the detail designs are prepared and submitted by the contractor, reviewed by the PMC & PIU and approved by the PIU. However the PIU may consult for special features with the PMC before approval. The PIU/PMC may at their own stipulate issues referred to them. Preliminary designs and estimates for tendering purposes are prepared by PMC and reviewed by PIU
- For all item rate contracts the designs are prepared by PMC, reviewed and approved by PIU.

Design control requirements are outlined below. The flow, handling and control of documents during design preparation and review are discussed in Section 10 of this Manual.

3.1 Design Preparation Process

Designs shall be prepared in accordance with the requirements of the Project, and applicable design standards and criteria, codes, specifications, and methodology. Detail designs prepared by contractors on turnkey contracts shall be based on the agreed design specifications.

For design calculations performed by computer, the design engineer shall verify design inputs and check outputs for reasonableness and compliance to requirements. Calculations prepared by hand shall be rechecked. The design engineer shall check the prepared designs and documents for completeness, correctness and legibility.

Design documents and calculations shall be reviewed by a senior engineer to ensure that the design method is acceptable and in accordance with the contractual design criteria, codes and standards, that the inputs are correct and the outputs reasonable.

Drawings will generally be prepared by computer aided design methods. As with design calculations, the designer shall check drawings for completeness, correctness, legibility and conformance to the design calculations, design standards, codes and specifications. A senior engineer shall then review drawings.

A common discussion with PMC design engineers & engineers are held. The comments of PIU engineers are then suitably incorporated.

Design documents, calculations and drawings shall be signed and dated by PMC design engineer and submitted to PIU by the PMC Team Leader.

3.2 Design checking Review Process

The design checking reviewer (PIU) shall examine the design documents, and if he has comments, prepare comment sheets and mark up a copy of the documents. If designs need to be reviewed by more than one person or agency, the primary reviewer shall consider the comments of other reviewers, discuss them as needed, and incorporate them into the marked up copy of the document.

Each design document and drawing shall be reviewed for compliance with contract requirements and specifications, applicable standards, codes and criteria. If the reviewer questions the accuracy of calculations or dimensions established by the designer, he shall

ask the designer to verify them but shall not change them himself. The PIU will Provide a certificate for checking , reviewing and approved.

3.3 Design Review Checklist

A basic design review checklist is as follows:

- o Check document revision number and revision dates (if applicable).
- o Check that the designer and reviewer have signed the document.
- o Verify the list of reference drawings and sources of special information and, as necessary, refer to the listed documents.
- o Check the general notes for clarity and completeness.
- o Review contract plans, specifications, addenda and all approved change orders, and check that the document complies with them.
- o Check the accuracy and completeness of a representative set of calculations.
- o Consider the aspect of constructability.
- o Annotate the drawing, specification, addendum or other document as appropriate.
- o Return the documents to the manager of the reviewing organization for return to the designer.

Specific items to be checked depend on the type of facility being designed.

4. CONSTRUCTION QUALITY CONTROL – GENERAL

This section provides an overview of construction quality control activities, including testing and site inspection. Materials control requirements are presented in detail in Section 5, while specific testing and inspection requirements for each category of works are presented in Sections 6 to 9 of this Manual.

4.1 Introduction

Construction quality control (CQC) is intended to provide a comprehensive, common and consistent framework for quality control across various contract packages. CQC comprises two main elements of quality control:

- Testing
- Inspections

Testing control covers the type of tests to be carried out, frequency of testing and stage of testing. Inspection control covers the timing of inspections, what has to be inspected and the inspection procedures.

CQC should be affected at five stages:

- Input Materials and Equipment Components
- In-process Activities
- Stage Completion
- Interfacing (of special importance in water supply and drainage contract packages)
- Final Completion

4.2 Testing

Various site tests on materials and works are required to be carried out by the contractor during construction. A well-equipped and properly operating site test laboratory is an important element of the quality assurance plan. A checklist showing typical testing equipment to be provided in the contractor's site laboratory is presented as Table 4.1.

The contractor shall set-up the site laboratory at the start of his project and inform the PIU and the PMC for conducting inspections. Laboratory equipment shall be properly calibrated, and calibration certificates should be kept at the laboratory for review by PIU and PMC, as necessary. Specialized tests at outside laboratories shall only be undertaken with the prior approval of the PIU.

Tests should be performed in accordance with the contract documents, as described in Sections 5 through 8 of this Manual. The control of test reports shall be done as stipulated in Section 10 of this Manual. All test samples should be preserved, with proper identification numbers, test log reference, test date, and other applicable information until otherwise instructed by PMC/PIU. These samples must be stored on site by the contractor.

Table 4.1
Checklist of Site Testing Laboratory Equipment

Contract Package No: _____ Name of Work: _____ Contractor Name: _____

Testing Equipment	Type of Test	Yes (1)	No (1)
Balance (2 type) volume measuring apparatus & hand tools etc.	General laboratory test		
Oven	Drying and moisture content determination		
Sieves, sieve shaker and hydrometer	Grain size analysis and classification of soil and aggregates		
Atterberg limit apparatus	<i>Plasticity of Soil</i>		
Procter/modified proctor (IS heavy) density equipment	Soil compaction test		
Sand replacement cylinder and core cutter	In-situ density test for field compaction control		
Compression testing machine 100 MT and 5 MT capacity	Compressive strength of cement, concrete, brick, etc.		
Cylinder and cube moulds	Concrete sampling		
Slump test	Workability and control of water in concrete casting		
Vicat apparatus	Setting time determination of cement		
Laboratory CBR equipment	Determination of strength of road pavement and layers		
Dynamic cone penetration test equipment	Determination of strength of road pavement and layers		
SPT equipment and hand auger	Boring, sampling and soil strength test		
Unconfined compression testing machine	Determination of shear strength of cohesive soil		
Direct shear equipment	Determination of angle of internal friction of soil		
Consolidation test equipment	Determination of settlement /compressibility of soil		
Los Angeles Abrasion test/ Impact Testing Machine	Los Angeles Abrasion test/ Impact Test		
Core drilling equipment	In-situ sampling		
Bitumen test equipment	Ductility test and Penetration test		
Measuring instruments	Measurement of sizes		
Leak test equipment	Pipeline jointing		
Marshal stability test equipment	Road works. To be available at Hot Mix Plant site.		
Metallic Contact Digital Thermocouple	Checking temperature of bitumen		
Leveling instruments	Tolerances of roads surfaces		
Any Others, Specify			

Note: 1. Yes or No to be tick marked by PIU to indicate the equipment at the site laboratory)

In addition to tests performed on site, the contractor is responsible for specialized tests which are performed at approved laboratories, and for factory inspections and tests performed by manufacturers or third parties during the manufacturing of various materials and equipment components, as stipulated in the contract documents.

4.3 Site Inspections

Site inspections shall be carried out to ensure that the materials and construction activities conform to the prescribed standards. Site inspections can be divided into day-to-day supervision and periodic quality inspection. The suggestions in respect of these two have been elaborated herein.

4.3.1 Day-to-day supervision

The day-to-day site supervision of all construction activities shall be carried out by the PMC. This includes checking of lines, levels and layouts and on-site checks. Progress monitoring and expediting shall also be carried out by the PMC. The supervisory team of the PMC shall ensure that materials that have been rejected or for which a conformance report has not yet been issued are not used in works.

Construction equipment is a major component of quality assurance system. The equipment requirements have been laid out in the Contract documents. It is necessary that the PMC check the adequacy of the equipment used by the contractor for construction as per the prescribed standards and specifications. The equipment used for construction shall be recorded in the daily logs.

4.3.2 Periodic quality inspections

The Senior Construction Engineer PMC & PIU (engineer in charge) shall carry out periodic quality inspections during in-process, stage completion, interfacing and final completion, and during all critical activities as per the following examples:

- excavation
- formation of embankments
- placing of reinforcing steel
- concrete batching and pouring
- hot mix operations
- laying of pavement layers
- laying and jointing of pipes
- installation of electrical and mechanical equipment
- testing, trial runs and commissioning of electro-mechanical equipment and plants

The PIU and PMC shall also inspect the materials certified by manufacturers and materials and equipment components approved by third parties upon delivery to the site. The contractor shall give advance notice to the PIU and PMC when critical activities are proposed or major equipment items are to be delivered.

On completion of one stage of the construction and before proceeding to the next stage (such as from steel binding to concreting for RCC works) the PMC where necessary shall

inspect and certify the quality of the works completed before granting approval for the next stage of the works to start. The final inspection shall encompass tests on completion and trial runs. The certification of quality will be based on the documents and the periodic site visits. The PMC representative and the PIU representative should witness not less than 30% and 10% of the test conducted respectively.

4.3.3 Squad checks

The concept of Squad Checks has been adopted to have an external review of quality of works executed. The squad checks should be conducted jointly by the PIU, PIU, and PMC. A fixed timetable is not suggested for this. The tentative agenda for the squad checks is described as follows:

- Physical inspection of the works under execution and inspection of quality of workmanship;
- review of site documentation and contractor compliance;
- sample verification of test reports and quality certificates;
- review of issues, constraints and lacunae in quality system implementation;
- preparing of action plans for improving the quality; and
- performance appraisal of the contractors.

Formats for recording minutes of progress review meetings and for recording interim evaluations of contractor's performance are included in Appendix C, as Format F-16 and F-17 respectively.

4.4 Quality Certification and Acceptance

The PMC shall be responsible to certify that the items included in the contractor's Interim Payment Certificate satisfy the required quality of works and are acceptable with regard to the specifications and standards prescribed under the contract before the progress bill is passed for payment. PIU should signify acceptance of the PMC's quality certification by countersigning it. A format for this quality certification and acceptance is included in Appendix C, as Format F-18.

5. CONTROL OF MATERIALS AND EQUIPMENT COMPONENTS

This section provides an overview of control requirements for materials and equipment components, including site testing, manufacturers' certification and third party inspection.

5.1 General

Control and approval of construction materials and equipment components to be incorporated in the works shall be based on the following:

- Test reports for materials tested at site, such as cement, sand, water, and aggregates. The contractor will perform all tests. The designated PMC representative & PIU representative shall witness at least 30% & 10% of tests respectively. They shall sign the report in token of witnessing.
- Manufacturer's certificates and IS mark for manufactured items indicated in table 5.3 or as stipulated in the contract; and
- Third party inspection for various items as per contract documents.

5.2 Materials Tested on Site

The materials to be tested on site include cement, water, aggregates for concrete, bricks and stones, soil for embankments, and aggregates and bituminous materials for road works. For aggregates and soil, the contractor shall obtain the approval of the borrow source or quarry from the PIU and the PMC before extracting material. The list of materials to be tested on site is given in Table 5.1 below. Test procedures are presented in Table 5.2, under the referenced procedure numbers. Test report formats are included in Appendix A. The reports are to be maintained in a bound register, where in 3 copies of report will be prepared, two copies to be submitted with monthly report to PMC & PIU and third copy to be retained by contractor.

Table 5.1
List of Materials Tested on Site

Sl. No.	Material	Test Procedure No.
1	Cement	QC-M-01
2	Sand	QC-M-02
3	Water for Construction Works	QC-M-03
4	Bricks	QC-M-04
5	Size Stone	QC-M-05
6	Coarse Aggregate for Concrete Work	QC-M-06
7	Soil/Earth/sub-grade material	QC-M-07
8	Granular Sub-base (GSB) Material	QC-M-08
9	Material for WBM / WMM	QC-M-09
10	Metal for BM/DBM/BC/Surface Dressing/MSS/Premix Carpet	QC-M-10
11	Binder for WBM	QC-M-11
12	Fine Aggregate for DBM/BC	QC-M-12
13	Lime	QC-M-13
14	Bitumen	QC-M-14
15	Borrow Material	QC-M-15

16	Steel (to be procured directly from manufacturer along with his test certificates) This shall also be got tested in local authorized test laboratory (by EE-PIU).	
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Table 5.2
Procedures for Testing Materials on Site

CEMENT			QC-M-01	
Sl. No.	Type of Test	Test Report Format No.	Frequency of Test	Timing of Test/ Inspection
1	Normal consistency	TC-M-01-01	One for each source and when called for by the Engineer	On receipt of material at site and before using as directed by the Engineer. Test certificate to be produced to the Engineer before use.
2	Fineness	TC-M-01-01		
3	Setting time – Initial / final	TC-M-01-01		
4	Compressive strength - 72 hrs, 168 hrs, 672 hrs.	TC-M-01-01		
For sulphate resistant cement as per IS-12330 OPC 43/53 shall conform to IS 8112/ 12269 and both 56 and 90 days strength shall be tested.				

SAND			QC-M-02	
Sl. No.	Type of Test	Test Report Format No.	Frequency of Test	Timing of Test/ Inspection
1	Sieve analysis	TC-M-02-01	One test for 15 m ³	On receipt at site and test certificate to be produced to the Engineer before use.
2	Fineness modulus	TC-M-02-01	One test for 15 m ³	
3	Deleterious constituents	TC-M-02-01	One test for 15 m ³	
4	Bulking test	TC-M-02-01	One test per Source	

WATER FOR CONSTRUCTION WORKS			QC-M-03	
Sl. No.	Type of Test	Test Report Format No.	Frequency of Test	Timing of Test/ Inspection
1	Alkalinity and acidity as per IS-3025	TC-M-03-01	Once per source of supply and when called for by the Engineer	Before use of water from that source
2	Solids	TC-M-03-01		

BRICKS			QC-M-04	
Sl. No.	Type of Test	Test Report Format No.	Frequency of Test	Timing of Test/ Inspection
1	Compressive strength	TC-M-04-01	One test per 50,000 bricks or part thereof	On receipt at site
2	Physical properties	TC-M-04-01		
3	Water absorption test	TC-M-04-01		

SIZE STONE			QC-M-05	
Sl. No.	Type of Test	Test Report Format No.	Frequency of Test	Timing of Test/ Inspection
1	Water absorption test	TC-M-05-01	One test per source and when called for	On receipt at site
2	Dimension check	Lab format	As directed by the Engineer	
3	Type of rock	Lab format		

COARSE AGGREGATE FOR CONCRETE			QC-M-06	
Sl. No.	Type of Test	Test Report Format No.	Frequency of Test	Timing of Test/ Inspection
1	Aggregate Impact or Los Angeles Abrasion Value as per IS-2386 Part-IV	TC-M-06-01/1 TC-M-06-01/2	One for each source of supply and when called for by the Engineer	On receipt of material at site
2	Soundness as per IS-2386 Part-V	TC-M-06-02		
3	Alkali Aggregate Reactivity as per IS-2386 Part-IV	Lab Format		
4	Flakiness Index	TC-M-06-03		
5	Gradation by wet sieve analysis	TC-M-06-04		
6	Water Absorption	TC-M-05-01		
When required, the contractor shall furnish the mix design along with material properties at least 15 days in advance.				

SOIL/EARTH/SUB-GRADE MATERIAL			QC-M-07	
Sl. No.	Type of Test	Test Report Format No.	Frequency of Test	Timing of Test/ Inspection
1.	Swelling index IS 2720 part XL	TC-M-09-01	Two sets for 3000 m ³ or part thereof	On receipt at site
2.	Liquid limits and plasticity index	TC-M-09-02		
3.	Deleterious material IS 1498	Lab format		
4.	OMC & MDD Test	TC-M-09-03		
5.	Chemical properties	Lab format		
6.	Grain Size Distribution Graph (by wet sieve analysis)	TC-M-09-04		
7.	Void ratio gradation	Lab format		
8.	Soaked CBR test (optional)	TC-M-07-01	Two sets for 3000 m ³ or part thereof and as directed by the Engineer	

GRANULAR SUB-BASE MATERIAL			QC-M-08	
Sl. No.	Type of Test	Test Report Format No.	Frequency of Test	Timing of Test/ Inspection
1	California Bearing Ratio Test	TC-M-07-01	As required	On receipt at site
2	Material combinations	Daily log		
3	Moisture content as per IS-2270	TC-M-07-02	1 test per 250 m3 or part thereof	Prior to compaction
4	Fineness value BS 812 Part III	Lab format	As required	On receipt at site
5	Soundness of material	TC-M-06-02		
6	Air voids content	Lab format		
7	Gradation by wet sieve analysis	TC-M-06-04	1 test per 200 m3 or part thereof	
8	Atterberg limits	TC-M-09-02		
9	Deleterious constituents	Lab format		
10	OMC and MDD	TC-M-09-03		
The contractor shall furnish the GSB design mix along with material properties and test results at least 15 days before laying GSB at site.				

MATERIAL FOR WBM / WMM			QC-M-09	
Sl. No.	Type of Test	Test Report Format No.	Frequency of Test	Timing of Test/ Inspection
1	Aggregate Impact Value	TC-M-06-01/1	One test for 200 m ³	On receipt at site
2	Grading by wet sieve analysis	TC-M-06-04	One test for 100 m ³	
3	Flakiness Index and Elongation Index	TC-M-06-03	One test for 200 m ³ of aggregate	
4*	Atterberg limits of binding material *(Only for WBM)	TC-M-09-02	One test for 25m ³ of binding material	
5	Atterberg limits of portion of aggregate passing 425 micron sieve.	TC-M-09-02	One test for 100 m ³ of aggregate	
6	Water Absorption Test	TC-M-05-01	Initially one set of 3 representative specimen for each source of supply and subsequently, when warranted by changes in the quality of aggregate	
7	Soundness Test	TC-M-06-02	One for each source of supply and when called for by the Engineer	On receipt at site and when absorption value is more 2%
8*	Density of compacted layer of WMM *(Only for WMM)		One test for 500 m ³	

METAL FOR BM / DBM / BC / SURFACE DRESSING / MSS / PRE-MIX CARPET			QC-M-10	
Sl. No.	Type of Test	Test Report Format No.	Frequency of Test	Timing of Test/ Inspection
1	Aggregate Impact Value	TC-M-06-01/1	One test for 50 m ³ of aggregate or part thereof	On receipt at site and before using in the hot mixing
2	Flakiness Index and Elongation Index of aggregates	TC-M-06-03		
3	Water absorption of aggregates	TC-M-06-06	Initially one set of 3 representative	

METAL FOR BM / DBM / BC / SURFACE DRESSING / MSS / PRE-MIX CARPET			QC-M-10	
Sl. No.	Type of Test	Test Report Format No.	Frequency of Test	Timing of Test/ Inspection
4	Stripping value	TC-M-11-01	specimen for each source of supply and subsequently, when warranted by changes in the quality of aggregate	
5	Gradation by wet sieve analysis	TC-M-06-04	As directed by the Engineer for individual component and for combined coarse, fine aggregate and filler.	
6	Soundness Test	TC-M-06-02	One for each source of supply and when called for by the Engineer	On receipt at site and when absorption value is more than 2%
For DBM and BC, the contractor shall furnish the material properties and proposed job mix formula at least 20 days in advance.				

BINDER FOR WBM			QC-M-11	
Sl. No.	Type of Test	Test Report Format No.	Frequency of Test	Timing of Test/ Inspection
1	Atterberg Limit Test	TC-M-09-02	One test for 100 m ³ of binding material	On receipt at site

FINE AGGREGATE FOR DBM/BC			QC-M-12	
Sl. No.	Type of Test	Test Report Format No.	Frequency of Test	Timing of Test/ Inspection
1	Passing 2.36 mm sieve and retained on 75 micron sieve	Daily log	As directed by the Engineer	Before use
2	Deleterious matter	Daily log	Visual observation of lot before use	
LIME			QC-M-13	
Sl. No.	Type of Test	Test Report Format No.	Frequency of Test	Timing of Test/ Inspection
1	Chemical properties as per IS: 6932, 1514	Lab format	3 final test samples for a lot size up to 100 tons as per Table 3 in IS 712-1984.	On receipt at site.
2	Physical properties as per IS: 6932	Lab format		

BITUMEN			QC-M-14	
Sl. No.	Type of Test	Test Report Format No.	Frequency of Test	Timing of Test/ Inspection
1	Grade of bitumen as directed/defined (Penetration Test)	TC-M-10-01	Two samples per test subject to all or some tests as directed by the Engineer	On receipt of material at site before unloading from the truck
2	Ductility Test	TC-M-10-02		
3	Flash and Fire Point Test	Lab format		
4	Viscosity Test	Lab format		
5	Softening Test	Lab format		

BORROW MATERIAL <i>(Soil to be used in Embankment / Subgrade / GSB)</i>			QC-M-15	
Sl. No.	Type of Test	Test Report Format No.	Frequency of Test	Timing of Test/ Inspection
1	Digging of borrow area for sampling	No format	25 m c/c or closer depending upon soil strata variation	Before material is extracted for use in construction.
2	Sand Content	TC-M-15-01 (use relevant test report formats and summarize results in TC-M-15-01)	2 sets of observation per 3000 m ³ of soil and in each 6 observations required.	Before material is extracted for use in construction.
3	Wet Sieve Analysis			
4	Plasticity Index			
5	Modified Proctor Density			
6	Soaked CBR Test			
7	OMC			
8	MDD			
Borrow material source must be approved before extracting material.				

5.3 Materials And Equipment Certified By Manufacturer

Acceptance of certain manufactured materials and equipment components, as stipulated in the contract, shall be based on test certificate(s) from the manufacturer conforming to IS and on visual inspection. These items shall bear the IS mark. PIU and PMC shall review the manufacturers' certificates for conformance to contract requirements before these items are delivered to the site. Upon their delivery and before their installation or otherwise incorporation in the works PIU and PMC shall inspect the condition of these items. Inspection criteria shall be decided jointly by PIU and PMC. They may decide to have the material additionally tested in Laboratory. The cost of such tests will be borne by the contractor. Materials and equipment subject to manufacturer's certification are listed in Table 5.3.

Table 5.3**List of Materials and Equipment Certified by Manufacturer**

- Steel/Reinforcing Steel
- Paint, Primers and Protective Coatings
- Glazing
- Water Proofing Compound
- GI, CI and PVC Pipes for general civil works
- Glazed Stoneware Pipes (GSW) for general civil works
- Gratings & Plates
- Manhole Covers
- Sanitary Fittings
- Metal Works such as windows, barbed wire, MS ladder, footrest, rolling shutters, etc.
- Joint Filler Material
- Pre-fabricated Water Tanks
- Traffic Signs
- Flow Measuring Devices – General
- Foot Rests
- Electrical Conduits
- Electrical Wires/Cables
- Switches & Sockets
- Distribution Boards
- Lights, Fans and Fixtures
- Earthing Material
- Insulators
- Lightning Arrestor
- Batteries
- Cable Termination Kit
- Fire Fighting Equipment
- Reduction Gearboxes
- Level Indicator & Controllers
- Laboratory Equipment
- Electrical Poles
- All other items as specified in the contract documents

5.4 Materials And Equipment Inspected By Third Party

Materials and equipment to be inspected by a third party vary from package to package, as stipulated in the contract documents. Third party inspection would normally take place at the factory during or upon completion of manufacture. Before site delivery, PIU and PMC shall review the third party inspection certificates for conformance to requirements. Upon delivery and before installation or incorporation in the works, PIU and PMC shall inspect the physical condition of these items and, if necessary, test them on site. Inspection criteria should be stipulated in the contract document. A list of materials and equipment suggested for inspection by third party is given in Table 5.4.

Table 5.4

List of Materials and Equipment Inspected by Third Party

- Flow Measuring Devices – Special
- Cranes & Lifting Tackles
- Electrical Cables – Special
- Butterfly Valves
- Sluice Valves
- Reflux Valves
- Air Valves
- Control Valves
- Vertical Turbine Pumps/Other Pumps
- Vertical Motors/Other Motors
- Gauges
- Electrical Starters
- Power Transformers
- Voltage Transformers
- Current Transformers
- Bus Ducts
- Switch Boards (HV/MV/LV)
- Battery Chargers
- DC Distribution Panel
- Steel Pipes – Lined & Coated
- Pre-stressed Concrete (PSC) Pipes
- NP Pipes for Sewers
- GSW pipes for sewers
- DI, CI and PVC pipes
- All specials and fittings for Water Treatment Systems
- In-situ Welding of Pipes

- In-situ Lining of Pipes
- Flash Mixer for WTP
- Aerator
- Clarifier
- Filter
- Sludge Treatment
- Motor Control Centre
- Indication-cum-Enunciation Panel
- Capacitors
- Sluice Gates
- Cable Trays
- All other items as specified in the contract documents

6. CONTROL OF GENERAL CIVIL AND STRUCTURAL WORKS

This section of the QA/QC Manual covers the testing of works and the inspection of workmanship for general civil and structural works. The key elements to be inspected in these works are concreting, stone masonry, and brickwork and finishes. The requirements for testing and control of materials for these works are outlined in Section 5.

6.1 Construction Sequence and Control Flow Charts

Flow charts indicating the construction sequence and control points for cement concrete and mortar works are shown in Figure 6.1 and 6.2 respectively.

6.2 Testing of Works

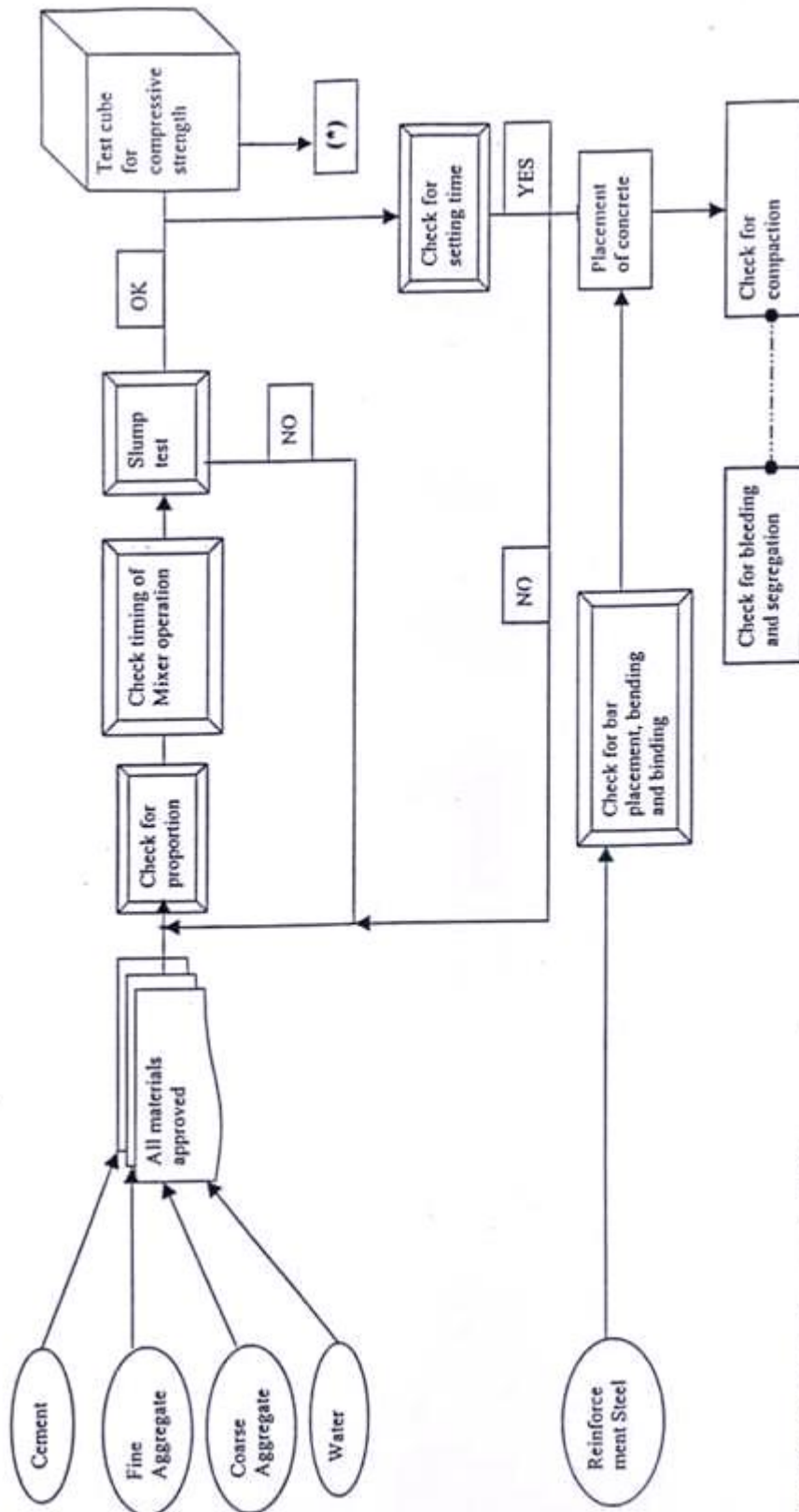
The works to be tested on site include excavation, cement concreting, and stone and brick masonry. All the materials proposed to be used in these works must have been tested by the contractor and approved by the PMC well in advance of these works. The contractor shall submit the concrete pouring report to the PIU and PMC as and when concreting is done, and shall obtain the approval of the PMC when a particular stage is completed and before proceeding to the next stage.

Tests for general civil and structural works are listed in Table 6.1. Test procedures are presented in Table 6.2, under the referenced test numbers. Required materials tests are also indicated (materials testing procedures are presented in Section 5). Test report formats are included in Appendix A. The contractor shall conduct tests as stipulated. The Representative of PMC and PIU will witness not less than 30% and 10 % of all the test conducted respectively.. This minimum percentage is not applicable where a higher % of this inspection is stipulated elsewhere.

6.3 Inspection Checklists

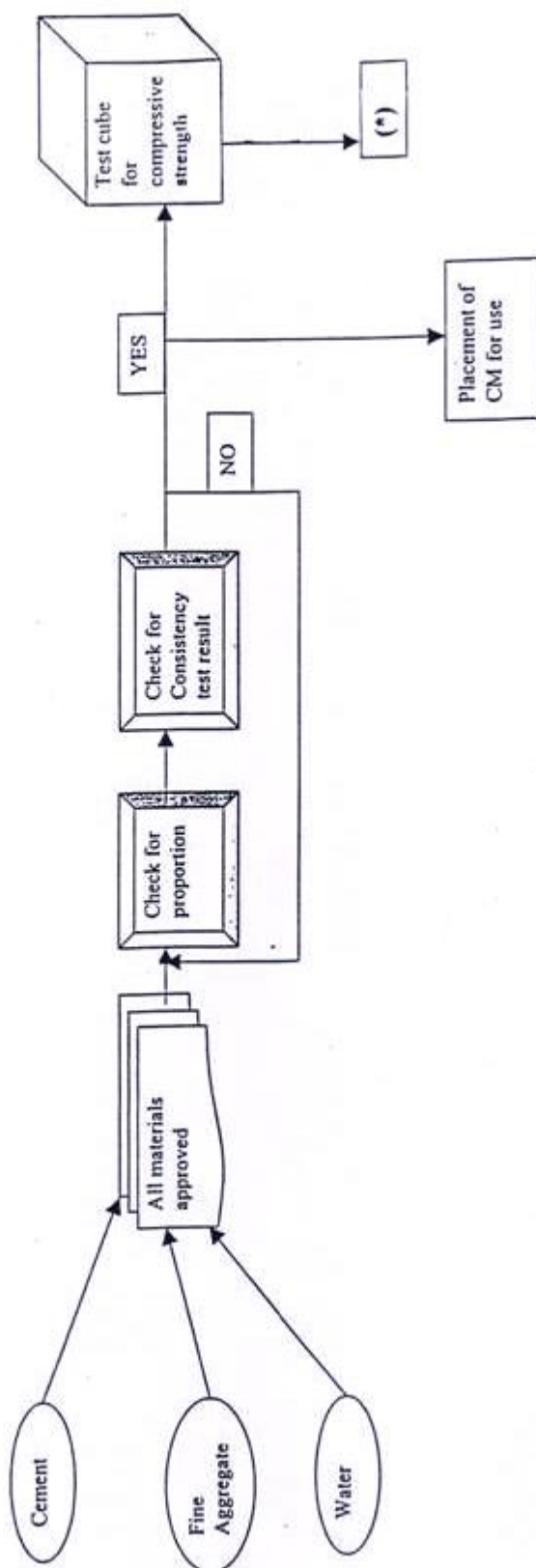
Inspection checklists for concreting, stone masonry, and brick masonry work and finishes, and building services and finishes are presented in Appendix B.

Figure 6.1
PROCESS CHART FOR PLAIN CEMENT CONCRETING/RCC WITH STAGES OF INSPECTION



Note (*): If test results are not satisfactory, corrective action shall be taken as directed by the Engineer.

Figure 6.2
PROCESS CHART FOR CEMENT MORTAR WITH STAGES OF INSPECTION



Note (*): If test results are not satisfactory, corrective action shall be taken as directed by the Engineer.

Table 6.1**List of Tests for General Civil and Structural Works**

Sl. No.	Process	Material		Test Ref. No.
		Name	Format No.	
1	Embankment Formation	Soil/Earth	QC-M-07	QC-G-01
2	Excavation/Backfilling			QC-G-02
3	Concreting	Steel	MC (1)	QC-G-03
		Cement	QC-M-01	
		Coarse Aggregates	QC-M-06	
		Sand	QC-M-02	
		Water	QC-M-03	
4	Size Stone Masonry	Size stone	QC-M-05	QC-G-04
		Cement	QC-M-01	
		Sand	QC-M-02	
		Water	QC-M-03	
5	Brick Masonry	Bricks	QC-M-04	QC-G-04
		Cement	QC-M-01	
		Sand	QC-M-02	
		Water	QC-M-03	

Note: 1. MC = manufacturer certified.

Table 6.2**Procedures for Testing General Civil and Structural Works**

Embankment Formation			QC-G-01	
Sl. No.	Type of Test	Test Report Format No.	Frequency of Test	Timing of Test/ Inspection
1	Moisture content as per IS-2720	TC-M-09-03	One test for each 250 m ³ of soil	In-process
2	Field density test as per IS-2720	TC-M-09-03	5-10 density tests for each 1000 m ² compacted area, or as directed by Engineer	
3	Compaction	Daily log	As per required number of passes	While compacting

Excavation/Backfilling			QC-G-02	
Sl. No.	Type of Test	Test Report Format No.	Frequency of Test	Timing of Test/ Inspection
1	Layout, slopes of excavation, benching and over-burden	Daily log	As directed by the Engineer	After excavation
2	Sub-soil water, shoring and strutting	Daily log		
3	Bottom levels and compaction	Daily log		
4	Soil classification	Daily log		
5	Backfilling and compaction	Daily log		After backfilling

Concreting			QC-G-03	
Sl. No.	Type of Test	Test Report Format No.	Frequency of Test	Timing of Test/ Inspection
1	Compressive strength as per IS-516	TC-G-01-01	- One test for 1-5 m³ of concrete - Two tests for 6-15 m³ of concrete - Three tests for 16-30 m³ of concrete - Four tests for 31-50 m³ + one set every 50 m³ of additional concrete work.	Test samples to be taken while pouring. Testing to be done as specified in contract.
2	Slump test per IS-1199	TC-G-01-02	Random checks throughout concreting as directed by the Engineer	Before pouring concrete
3	Inspection of steel reinforcement placement and bending, and formwork	Daily log	Before pouring concrete	Before pouring concrete
4	Concrete Pour Report	TC-G-01-03	When pouring is done	Immediately after pouring

Mortar			QC-G-04	
Sl. No.	Type of Test	Test Report Format No.	Frequency of Test	Timing of Test/ Inspection
1	Compressive strength as per IS-2250	TC-G-01-01	One sample for every 2 m ³ of mortar subject to a minimum of three samples for a day's work	Test samples to be taken while before mortaring. Testing to be done as specified in contract.
2	Consistency as per IS-2250	TC-G-02-01		

7. CONTROL OF ROAD WORKS

This section of the QA/QC Manual covers the testing of works and the inspection of workmanship for road works, including earthworks, placement of sub-base and WBM layers, application of prime and tack coats, and placement of bituminous layers. The requirements for testing and control of materials for road works are outlined in Section 5.

7.1 Construction Sequence and Control Flow Charts

Flow charts indicating the construction sequence and control points for road works are shown in Figures 7.1 to 7.9.

7.2 Testing of Works

The works to be tested on site include earthworks, placement of granular sub-base and WBM layers, application of prime and tack coats, and placement of bituminous layers. All the materials proposed to be used in these works must have been tested by the contractor and approved by the PMC well in advance of the works. The contractor shall obtain the approval of the PMC when a particular stage is completed and before proceeding to the next stage. Surface regularity and alignments shall be checked by leveling instrument.

Hot mix designs shall be submitted by the contractor to PIU and PMC for review and approval well before the planned start of hot mix operations. The hot mix plant shall be inspected by PIU and PMC and approved by PIU before commencing operations. Temperature tests on bitumen shall be carried out at the hot mix plant before delivery to the site, and immediately before placing and after compaction. Temperature tests shall be carried out by using metal contact digital thermocouple based temperature measuring device. The Contractor shall provide such devices as part of his site laboratory, and in sufficient quantity so that all required testing can be carried out as-and-when required. The contractor shall take the temperature readings in the presence of the PMC, and shall submit his test reports on a daily basis.

Tests for road works are listed in Table 7.1. Test procedures are presented in Table 7.2, under the referenced test numbers. Required materials tests are also indicated (materials testing procedures are presented in Section 5). Test report formats are included in Appendix A.

7.3 Inspection Checklists

An inspection checklist for road works is presented in Appendix B.

Figure 7.1
FLOW CHART FOR THE CONSTRUCTION OF EMBANKMENT AND SUBGRADE

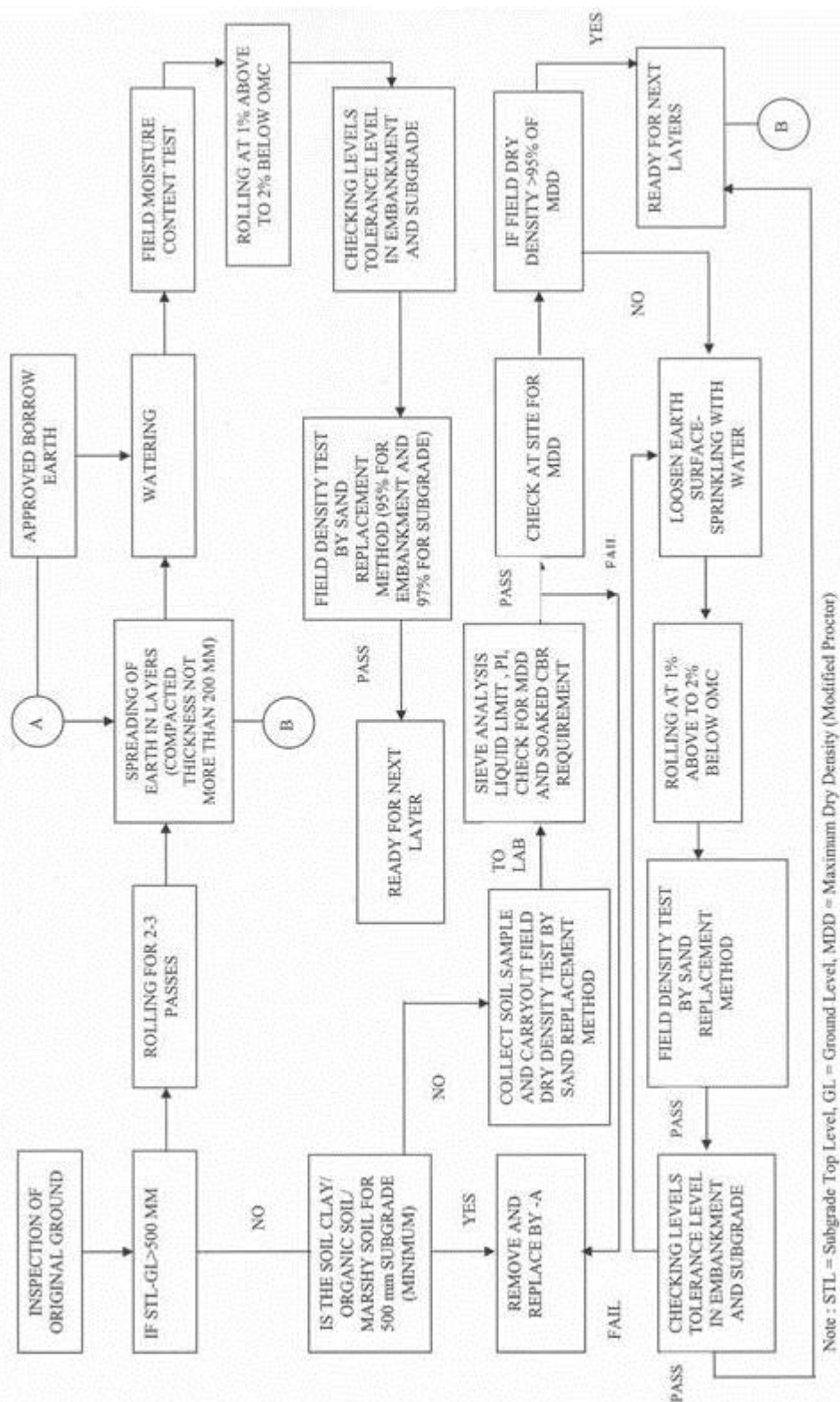
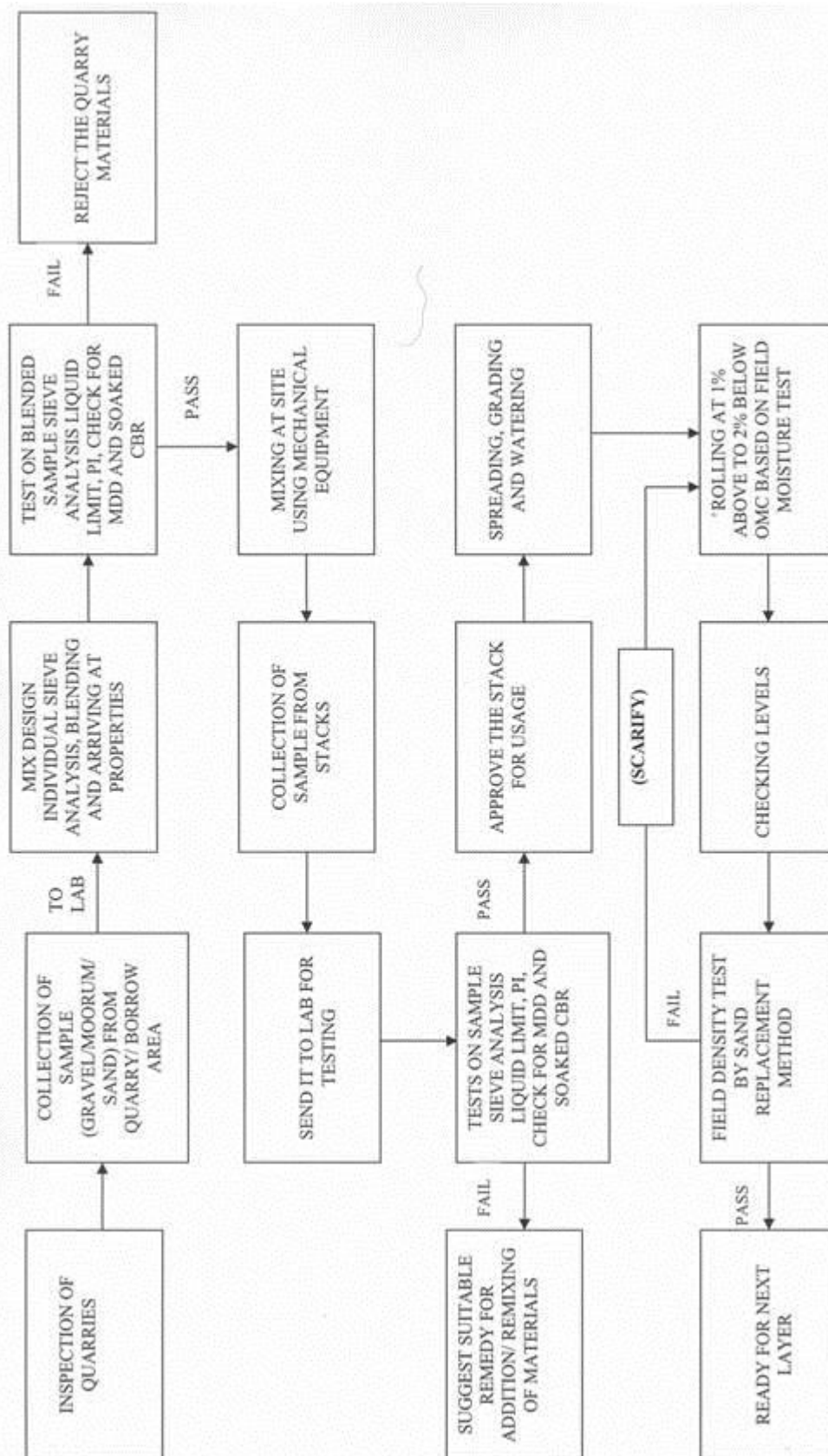


Figure 7.2
FLOW CHART FOR THE CONSTRUCTION OF GRANULAR SUBBASE



Note : MDD = Maximum Dry Density (Modified Proctor)

Figure 7.6
FLOW CHART FOR BITUMINOUS MIX DESIGN (BC)

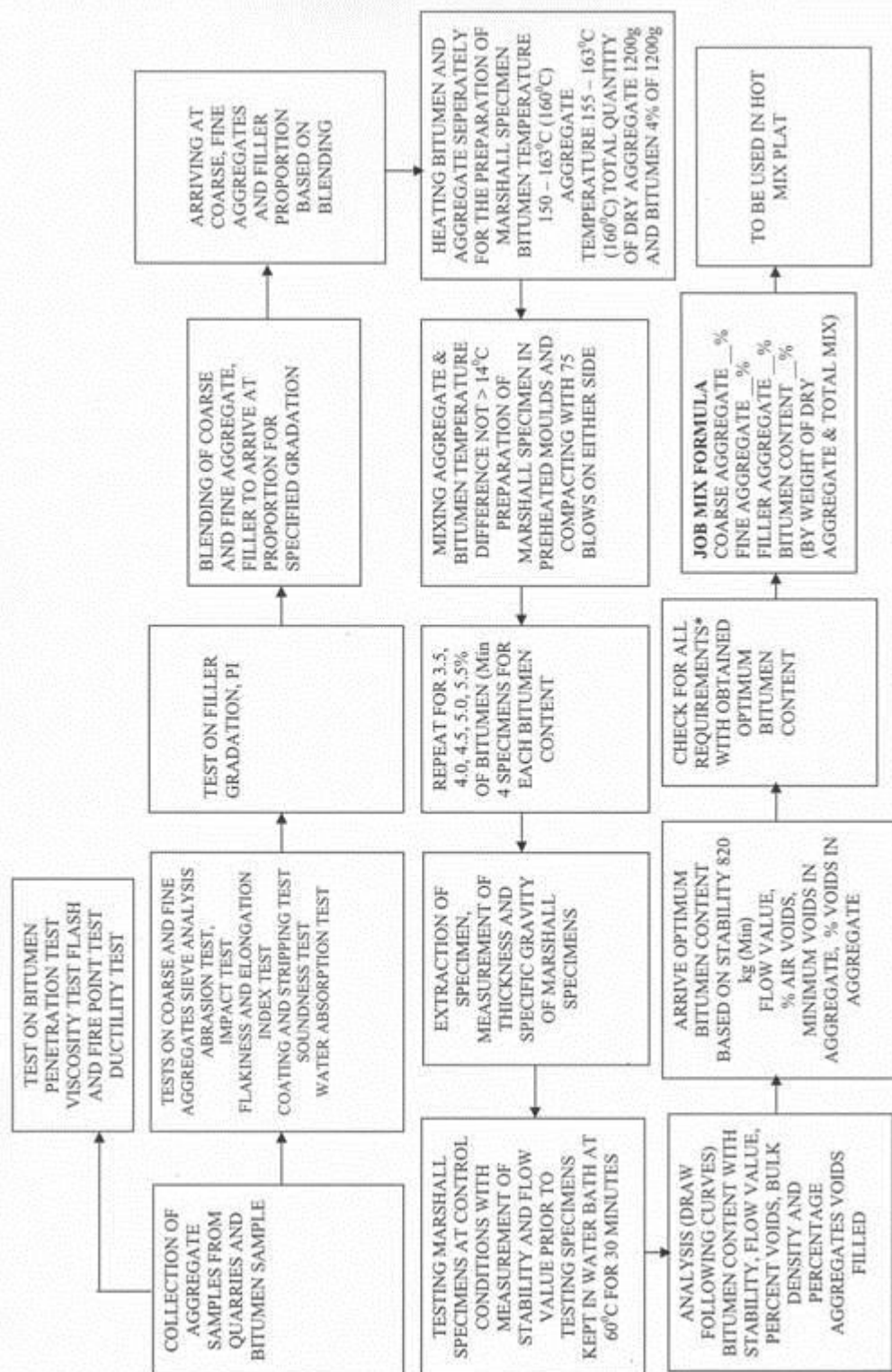


Figure 7.7
FLOW CHART FOR INSTALLATION AND RUNNING OF HOT MIX PLANT

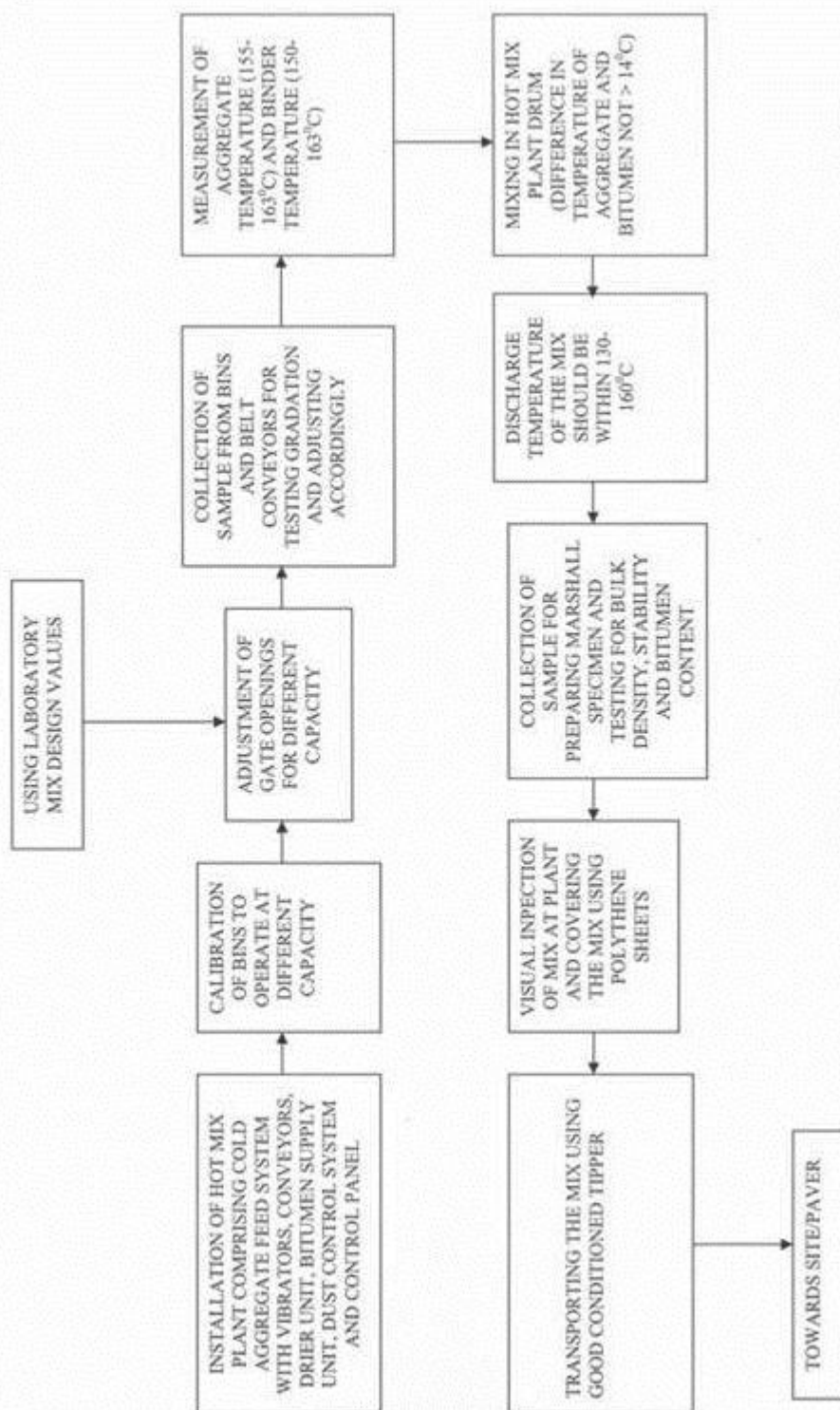
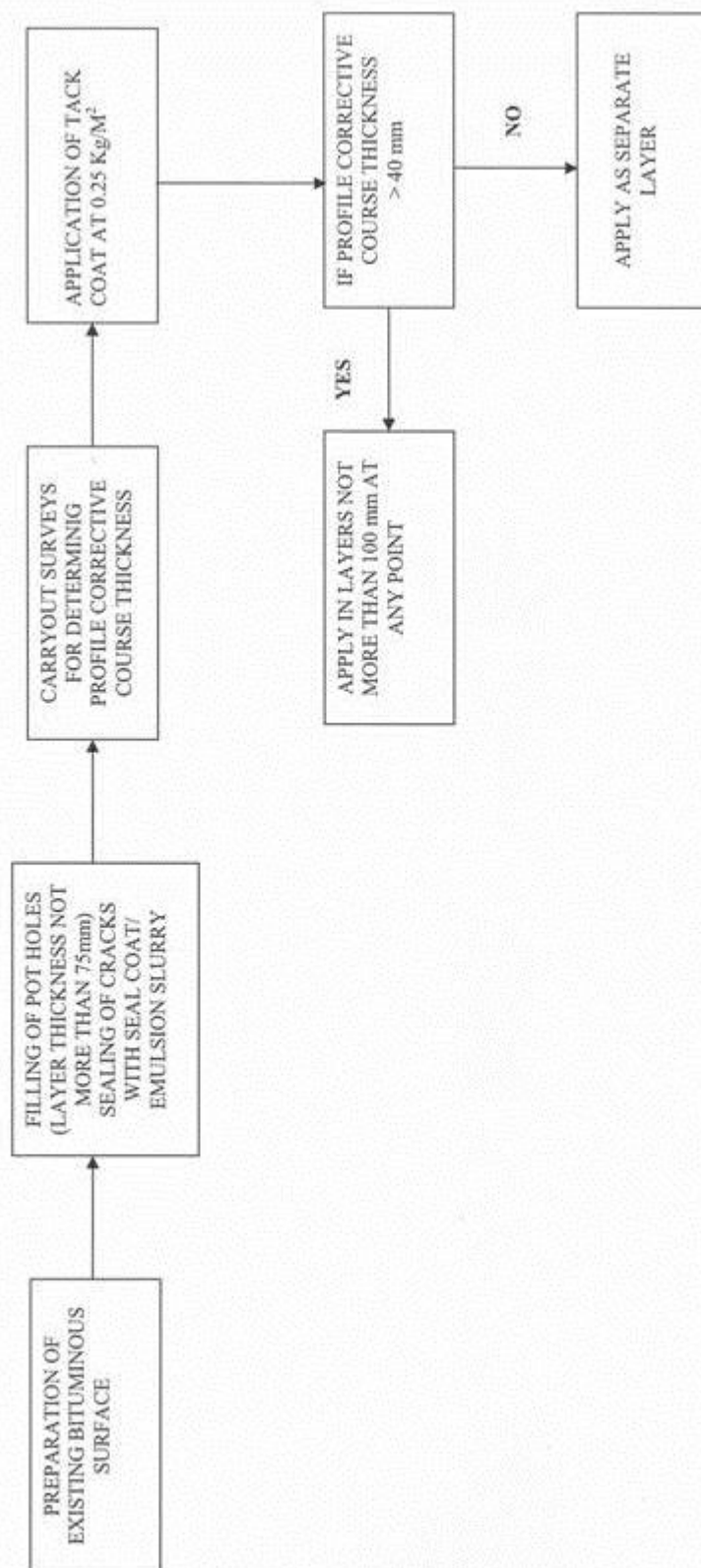


Figure 7.8
FLOW CHART FOR LAYING PROFILE CORRECTIVE COURSE



Note : PCC = Profile Corrective Course

Table 7.1

Figure 7.9
FLOW CHART FOR CONSTRUCTION OF DBM/BC LAYERS

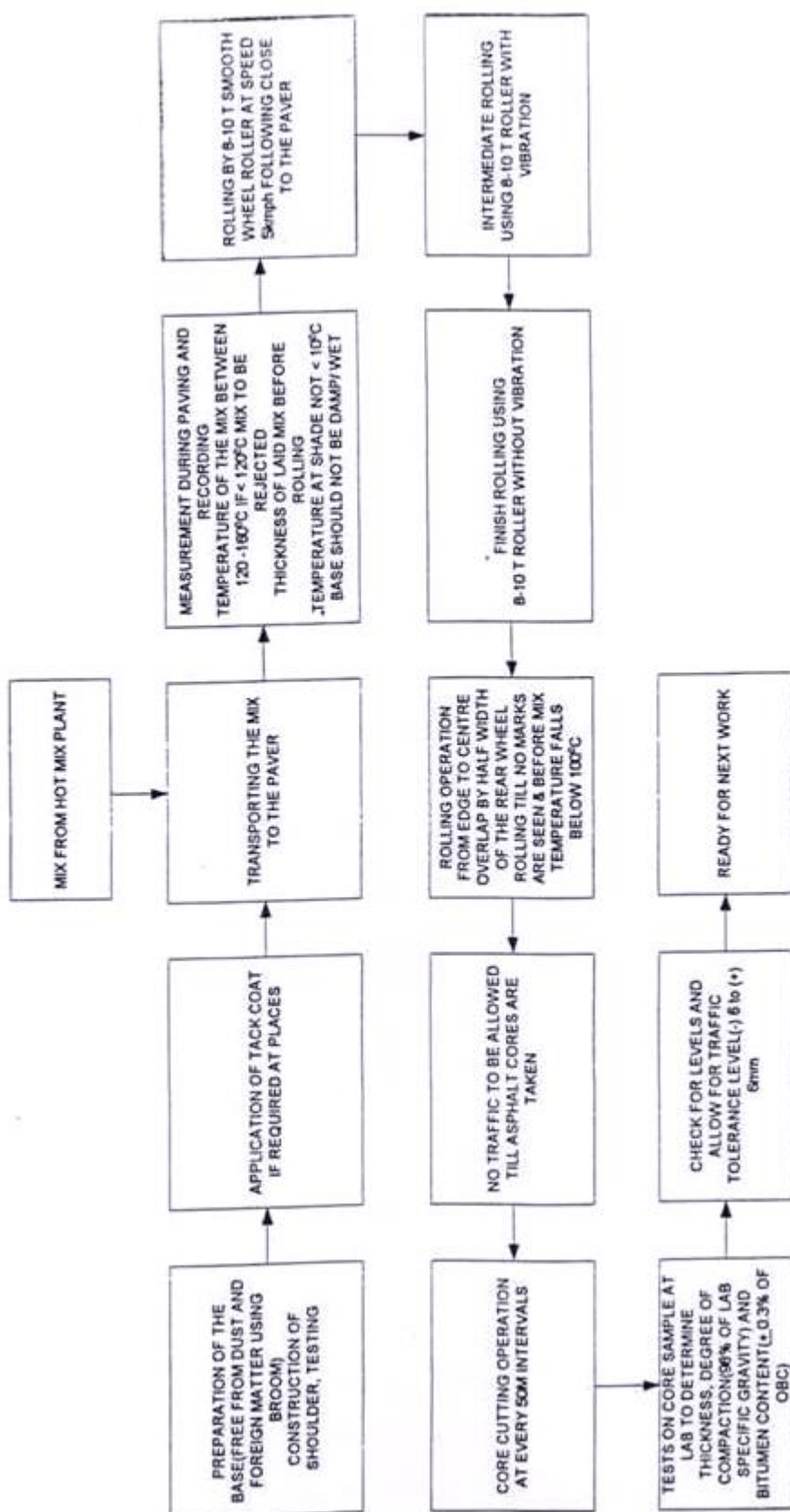


Table 7.1
List of Tests for Road Works

Sl. No.	Process	Material		Test Ref. No.
		Name	Format No.	
1.	Embankment Formation	Soil/Earth	QC-M-07 QC-M-15	QC-R-01
2.	Excavation			QC-R-02
3.	Granular Sub-base Laying	Granular Sub-base	QC-M-08	QC-R-03
4.	WBM Laying	WBM	QC-M-09	QC-R-04
		Binder	QC-M-11	
5.	Prime Coat Application	Bitumen	QC-M-14	QC-R-05
6.	Tack Coat Application	Bitumen	QC-M-14	QC-R-05
7.	Surface Dressing	Bitumen	QC-M-14	QC-R-06
		Metal	QC-M-10	
8.	Bituminous Macadam Laying	Bitumen	QC-M-14	QC-R-07
		Metal	QC-M-10	
9.	DBM Laying	Bitumen	QC-M-14	QC-R-08
		Metal	QC-M-10	
		Fine Aggregates	QC-M-12	
		Filler (Lime)	QC-M-13	
10.	Mix Seal Surfacing	Bitumen	QC-M-14	QC-R-06
		Metal	QC-M-10	
		Fine Aggregates	QC-M-12	
11.	Bituminous Concrete Laying	Bitumen	QC-M-14	QC-R-08
		Metal	QC-M-10	
		Fine Aggregates	QC-M-12	
		Filler (Lime)	QC-M-13	
12.	On Completion of Laying of BM / DBM / BC	Stage completion test		QC-R-09

Table 7.2
Procedures for Testing Road Works

Embankment Formation			QC-R-01	
Sl. No.	Type of Test	Test Report Format No.	Frequency of Test	Timing of Test/ Inspection
1	Moisture content as per IS-2720	TC-M-09-03	One test for each 250 m ³ of soil	In-process
2	Field density test as per IS-2720	TC-M-09-03	- For earthwork in embankment/cutting 5-10 density tests for each 1000 m² compacted area - For earthwork in sub grade/GSB and shoulders, 10 density tests for 500 m² compacted area	
3	Rolling operation	Daily log	As per required number of passes	While rolling

Excavation			QC-R-02	
Sl. No.	Type of Test	Test Report Format No.	Frequency of Test	Timing of Test/ Inspection
1	Layout, slopes of excavation, benching and over-burden	Daily log	As directed by the Engineer	After excavation
2	Sub-soil water, shoring and strutting	Daily log		
3	Bottom levels and compaction	Daily log		
4	Soil classification	Daily log		

Granular Sub-Base Laying			QC-R-03	
Sl. No.	Type of Test	Test Report Format No.	Frequency of Test	Timing of Test/ Inspection
1	Moisture content as per IS-2720	TC-M-09-03	One test for 500m ² of compacted soil (3 observations per test)	In-process
2	Field density test as per IS-2720	TC-M-09-03	10 observations selected randomly for every 500m ² of compacted area	
3	Rolling operation	Daily log	Required No. of passes	While rolling

WBM Laying			QC-R-04	
Sl. No.	Type of Test	Test Report Format No.	Frequency of Test	Timing of Test/ Inspection
1	Field Density Test by sand replacement method	TC-R-06-01	As directed by the Engineer	In-process

Prime Coat/Tack Coat Application			QC-R-05	
Sl. No.	Type of Test	Test Report Format No.	Frequency of Test	Timing of Test/ Inspection
1	Temperature Test	Daily log	At regular close intervals	In-process
2	Rate of spreading	TC-R-02-01	Three tests for every 50 m length	

Surface Dressing/Mix Seal Surfacing/ Pre-mix Carpet			QC-R-06	
Sl. No.	Type of Test	Test Report Format No.	Frequency of Test	Timing of Test/ Inspection
1	Rate of spread of mix material	TC-R-02-01	One test for every 500 m ³ of mix with 6 observations	In-process

Bituminous Macadam Laying			QC-R-07	
Sl. No.	Type of Test	Test Report Format No.	Frequency of Test	Timing of Test/ Inspection
1	Temperature Test	Daily log	At regular close intervals	In-process
2	Rate of spread of mix material	TC-R-02-01	2-3 observations at every 10 m interval during paving	

DBM/BC Laying			QC-R-08	
Sl. No.	Type of Test	Test Report Format No.	Frequency of Test	Timing of Test/ Inspection
1	Temperature Test	Daily log	At regular close intervals	In-process
2	Rate of spread of mix material	TC-R-02-01		
3	Stability of Mix/Marshall Stability Test	TC-R-05-01	3 samples for each 400 tons of mix produced subject to a minimum of 2 tests per plant per day.	While hot mixing

On Completion of BM / DBM / BC Layers Stage Completion Test			QC-R-09	
Sl. No.	Type of Test	Test Report Format No.	Frequency of Test	Timing of Test/ Inspection
1	Core test for compacted layer (bitumen content, density and voids)	TC-R-07-01	One test for 250 m ² of compacted area	On completion of stage and before proceeding to next stage
2	Surface regularity and control of alignment	TC-R-07-02	One test for every 300 m of road length	

7.4 Guidelines for Pavement Layers

Guidelines for gradation and placement of WBM layer are given in Table 7.3 and 7.4, respectively; guidelines for application of prime and tack coats are given in Table 7.5; and requirements of bituminous mixes are given in Table 7.6. (These guidelines are provided for easy reference. The standards and specifications stipulated in the contract shall be adhered to.)

Table 7.3
Gradation for WBM Layers

Grade-II (Size: 63 mm to 45 mm)		Grade-III (Size: 53 mm to 22.4 mm)		Screening Material (Size: 11.2 mm)	
IS Sieve Designation (mm)	% by weight passing	IS Sieve Designation (mm)	% by weight passing	IS Sieve Designation (mm)	% by weight passing
90	100	63	100	11.2	100
63	90-100	53	95-100	5.6	90-100
53	25-75	45	65-90	0.18	15-35
45	0-15	22.4	0-10		
22.4	0-5	11.2	0-5		

Table 7.4
Guidelines for Placement of WBM Layers

Quantity Required for 10 Sq. m Area (Compacted thickness 75mm)		
Grading	Size (mm)	Loose Quantity (m³)
II	63-45	0.91-1.07
III	53-22.4	0.91-1.07
Screening Material for II	11.2	0.20-0.22
Screening Material for III	11.2	0.18-0.21
Binding Material (II/III)	---	0.06-0.09

Table 7.5
Guidelines for Prime/Tack Coat Application

Particulars	Standard Specified Rate (kg/m²)
Granular surface treated with primer	0.25-0.30
Granular base not primed	0.35-0.40
Normal bituminous surfaces	0.20-0.25

Table 7.6
Requirements of Bituminous Mixes

Minimum stability (kN at 60° C)	9.0
Minimum flow (mm)	2
Maximum flow (mm)	4
Compaction level (Number of blows)	75 blows on each of the two faces of the specimen
Percent air voids	3-6
Percent voids in mineral aggregate (VMA)	See table 7.6(a)
Percent voids filled with bitumen (VFB)	65-75

Table 7.6 (a)
Minimum percent Voids in Mineral Aggregate (VMA)

Nominal Maximum Particle size (mm)	Minimum VMA, percent Related to Design Air Voids, Per cent ²		
	3.0	4.0	5.0
9.5	14.0	15.0	16.0
12.5	13.0	14.0	15.0
19.0	12.0	13.0	14.0
25.0	11.0	12.0	13.0
37.5	10.0	11.0	12.0

- Notes: 1. The nominal maximum particle size is one size larger than the first sieve to retain more than 10 per cent.
2. Interpolate minimum voids in the mineral aggregate (VMA) for design air voids values between those listed.

7.5 Tolerances

Requirements for surface regularity and tolerances are given below. (These requirements are for easy reference; the standard and special technical specifications as per the contract must be referred.)

7.5.1 Horizontal alignment tolerances

The horizontal alignment with respect to the centerline of the carriageway shall have a tolerance of ± 10 mm at the edges of roadway and of ± 25 mm lower layers.

7.5.2 Surface levels tolerances

Surface level tolerances are shown in Table 7.7.

Table 7.7
Surface Level Tolerances

Type of Surface	Tolerance in Level Compared with Longitudinal and Cross Profile
Sub-grade	+ 20 mm/- 25 mm
Sub-base	+10 mm/- 20 mm
Base Course	
(a) Machine laid	± 10 mm
(b) Manually laid	± 15 mm
Wearing Course	
(a) Machine laid	± 6 mm
(b) Manually laid	± 10 mm
Cement concrete pavement	+5 mm -6 mm*

* This may not exceed -8 mm at 0-30 cm from the edges.

7.5.3 Surface regularity of pavement courses

The maximum allowable difference between the road surface and a straight line parallel with or at right angles to the centerline of the road at points shall be:

- For bituminous surface: 3 mm
- For GSB/base courses : 8 mm

The maximum permitted number of surface irregularities measured under a 3 m long straight edge at the middle of each traffic lane along a line parallel to the center line of the road shall be as shown in Table 7.8.

Table 7.8
Maximum Number of Surface Irregularities

Maximum Number of Surface Irregularities on Bituminous Road and Shoulders				
Irregularity	4 mm		7 mm	
Length (m)	300	75	300	75
National/State Highways	20	9	2	1
Town Roads	40	18	4	2

8. CONTROL OF PIPELINE WORKS

This section of the QA/QC Manual covers the testing of works and the inspection of workmanship for pipeline works (i.e. water and sewer lines) and liquid retaining structures. The requirements for testing and control of input materials are outlined in Section 5.

8.1 Construction Sequence and Control Flow Charts

Flow charts indicating the construction sequence and control points for materials used in pipeline work and for pipeline works are shown in Figure 8.1 and 8.2 respectively.

8.2 Testing of Works

The works to be tested on site include bedding for pipelines, pipeline laying and jointing, and hydrostatic, leakage and water tightness tests after completion. All the materials proposed to be used in these works must have been tested by the Contractor and approved by the PMC well in advance of commencing works. The contractor shall obtain the approval of the PMC when a particular stage is completed and before proceeding to the next stage.

Tests for pipeline works and liquid retaining structures are listed in Table 8.1. Test procedures are presented in Table 8.2, under the referenced test numbers. For excavation, back-filling and re-paving works refer to Sections 6 and 7 of this Manual. Required materials tests are also indicated (materials testing procedures are presented in Section 5). Test report formats are included in Appendix A.

8.3 Inspection Checklists

An inspection checklist for pipeline works is included in Appendix B.

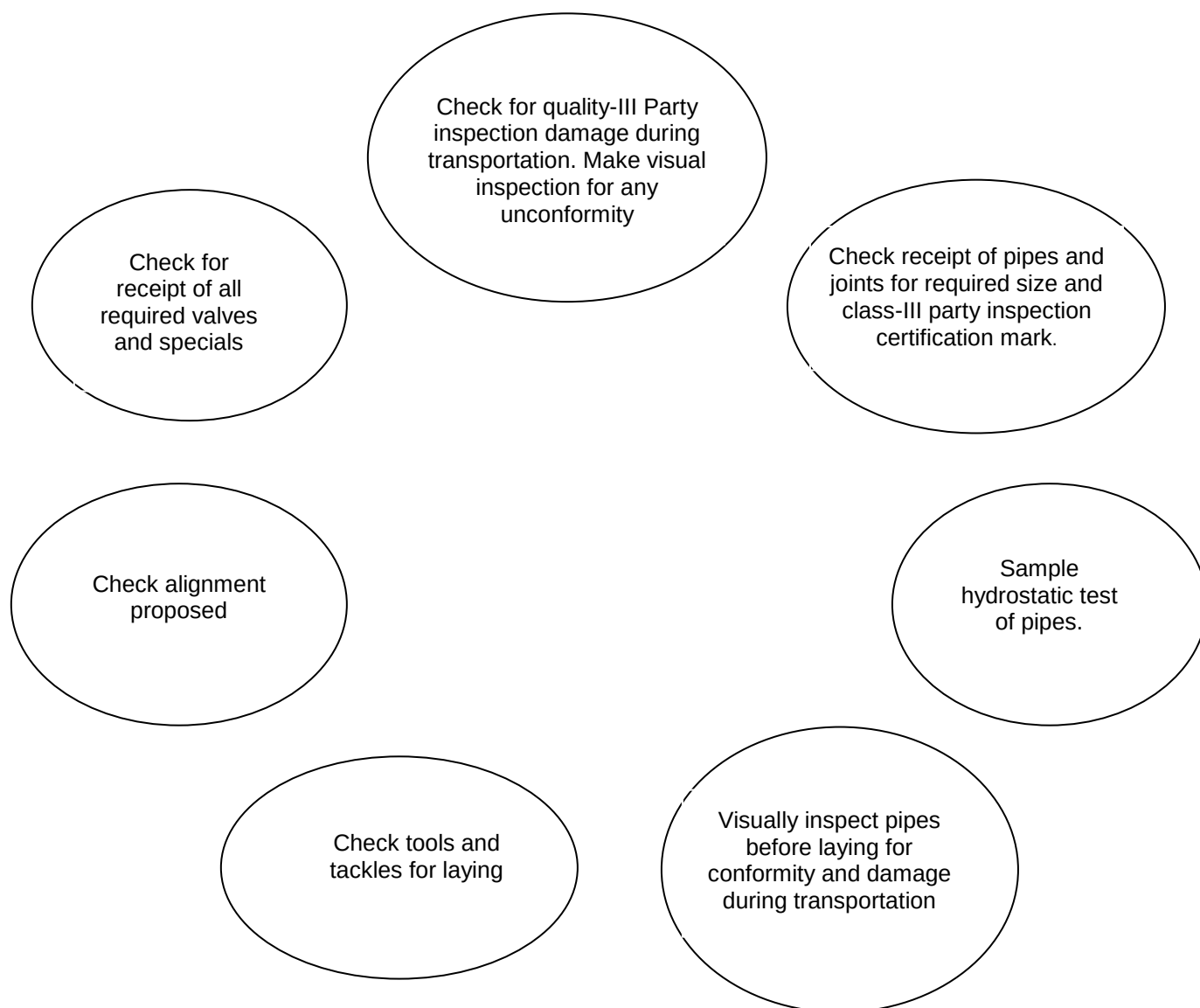
Figure 8.1 (b)**CHECKS FOR PREPARATION WORKS BEFORE LAYING WATER SUPPLY
SEWERAGE PIPE LINES**

Figure 8.2
PROCESS CHART FOR PIPELINE WORKS WITH STAGES OF INSPECTION

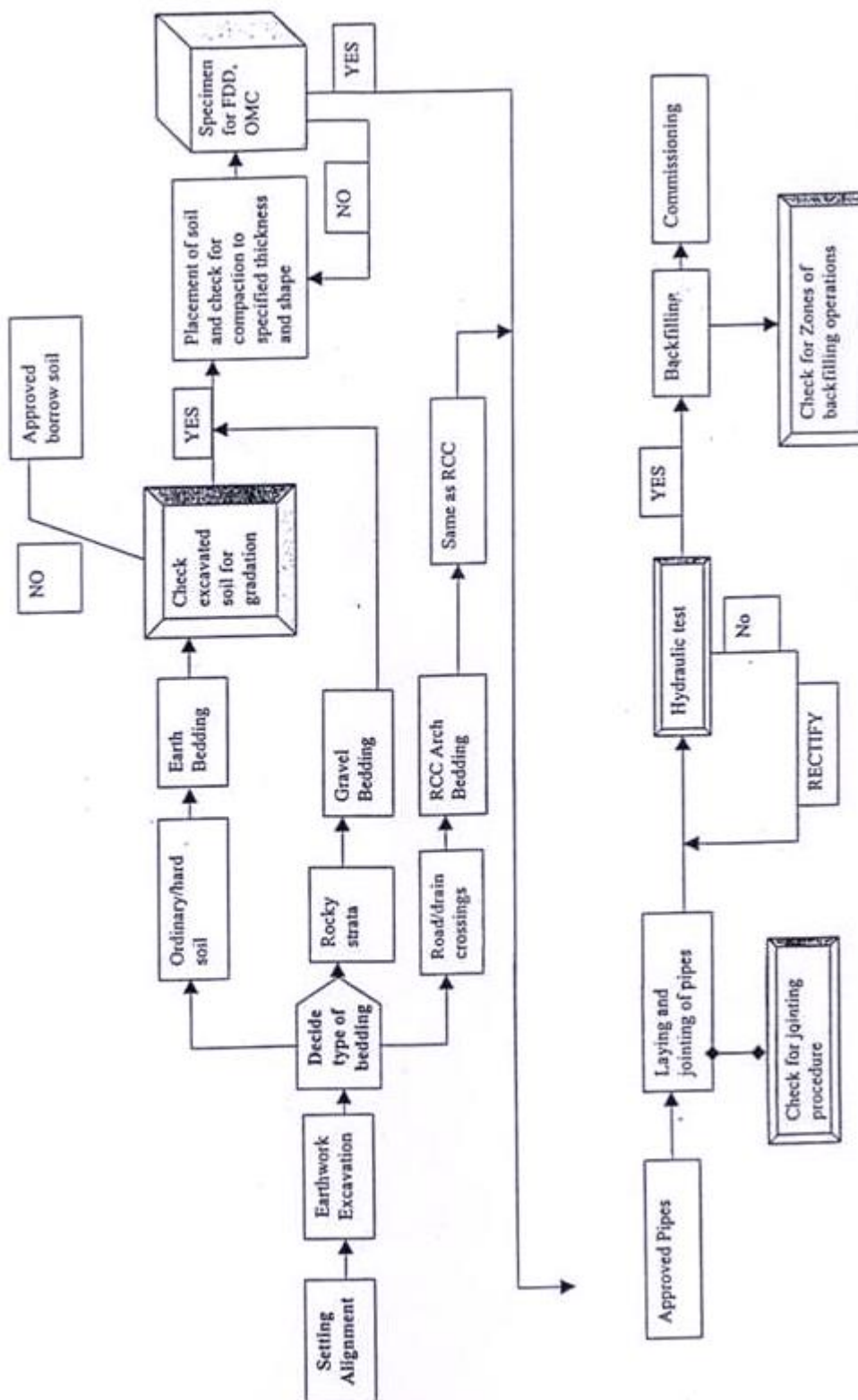


Table 8.1
List of Tests for Pipeline Works and Liquid Retaining Structures

Sl. No.	Activity	Material		Test Ref. No.
		Name	Format No.	
1	Bedding for Pipeline			
1.1	Earth Bedding	Earth/Soil	QC-M-07	QC-P-01
1.2	Gravel Bedding	Gravel	QC-M-08	
1.3	Concrete Bedding	Cement	QC-M-01	QC-P-02
		Sand	QC-M-02	
		Water	QC-M-03	
		Coarse Aggregate	QC-M-06	
		Steel	MC (1)	
2	Pipeline Laying and Jointing			
2.1	GSW Pipe	Cement	QC-M-01	QC-P-03
		Sand	QC-M-02	
		Water	QC-M-03	
		Pipes	MC (1)	
2.2	RCC and CI Pipes	Pipes	TPI/MC (1)	QC-P-02
		Gaskets	TPI/MC (1)	
2.3	Coated and Lined Steel/PSC Pipes	Coated and lined steel/ PSC pipes	TPI (1)	
3	Manhole/Valve Chamber Construction	Cement	QC-M-01	QC-P-02 QC-P-03
		Sand	QC-M-02	
		Water	QC-M-03	
		Bricks	QC-M-04	
		Coarse Aggregate	QC-M-06	
		Cover	TPI (1)	
		Frame/Vent shaft etc.	MC (1)	
		Steel	MC (1)	
4	Completion of Pipeline Laying and Jointing	Stage Completion Test		QC-P-04
5	Completion of Manhole/ Valve Chamber Construction	Stage Completion Test		QC-P-05
6	Completion of Liquid Retaining Structures (Wet Wells, Storage Reservoirs, Pretreatment Units, RCC Open Channels, etc.)	Stage Completion Test		QC-P-06

Note: 1. MC = manufacturer certified; TPI = third party inspection.

Table 8.2
Procedures for Testing Pipeline Works and Liquid Retaining Structures

Earth Bedding			QC-P-01	
Sl. No.	Type of Test	Test Report Format No.	Frequency of Test	Timing of Test/ Inspection
1	Moisture content as per IS-2720	TC-M-09-03	One test for each 250 m ³ of soil	In-process
2	Field density test as per IS-2720	TC-M-09-03	One test for each 100 m ² of compacted area	

Concreting			QC-P-02	
Sl. No.	Type of Test	Test Report Format No.	Frequency of Test	Timing of Test/ Inspection
1	Compressive strength as per IS-516	TC-G-01-01	- One test for 1-5 m³ of concrete - Two tests for 6-15 m³ of concrete - Three tests for 16-30 m³ of concrete - Four tests for 31-50 m³ of concrete + one set every 50 m³ of additional concrete work.	Test samples to be taken while pouring. Tests to be done as specified in the contract.
2	Slump test per IS-1199	TC-G-01-02	Random checks throughout concreting period as directed by the Engineer	Before pouring concrete
3	Steel reinforcement placement and bending	Daily log	Before pouring concrete	Before pouring concrete
4	Concrete Pour Report	TC-G-01-03	When pouring is done	Immediately after pouring

Mortar			QC-P-03	
Sl. No.	Type of Test	Test Report Format No.	Frequency of Test	Timing of Test/ Inspection
1	Compressive strength as per IS-2250	TC-G-01-01	One sample for every 2 m ³ of mortar subject	Test samples to be taken while

2	Consistency as per IS-2250	TC-G-01-02	to a minimum of three samples for a day's work	placing. Tests to be done as specified in the contract.
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Completion of Pipeline Laying and Jointing			QC-P-04	
Sl. No.	Type of Test	Test Report Format No.	Frequency of Test	Timing of Test/ Inspection
1	Hydrostatic test for NP pipes	TC-P-04-01	One test for defined stretch	On completion of stage
2	Hydrostatic test for pressure pipes	TC-P-04-02	One test for defined stretch	On completion of stage

Completion of Manhole/Valve Chamber			QC-P-05	
Sl. No.	Type of Test	Test Report Format No.	Frequency of Test	Timing of Test/ Inspection
1	Leakage Test	TC-P-05-01	100% inspection	On completion of stage

Completion of Liquid Retaining Structures			QC-P-06	
Sl. No.	Type of Test	Test Report Format No.	Frequency of Test	Timing of Test/ Inspection
1	Water tightness for underground structures	TC-P-06-01	One test per structure	On completion of stage
2	Water tightness for elevated structures	TC-P-06-02	One test per structure	

9. CONTROL OF ELECTROMECHANICAL WORKS

This section of the QA/QC Manual gives an overview of the quality control requirements for electromechanical works, such as water treatment and supply systems, sewage treatment plants, compost plants, pumping systems, and power supply and distribution systems. The requirements for testing and control of input materials and components, including manufacturers' certification and third party inspections, are outlined in Section 5.

Materials and components to be incorporated into electromechanical works shall be inspected by PIU and PMC as soon as they are delivered, to ensure that they meet the specifications and design requirements, are in agreement with shipping documentation, and are accompanied by manufacturer's certifications or third party inspection certificates, as applicable. Accepted materials and equipment shall be properly stored by the contractor until needed. If manufacturer's installation instructions conflict with design or contract requirements, the PIU, PIU and shall be notified immediately. Installation shall proceed only after the conflict has been resolved.

A series of inspections and tests during installation and completion of electromechanical works shall be performed by the contractor or the equipment manufacturer and witnessed by PIU and PMC, as follows:

- **Preparatory Inspections:** Prior to installation, the civil and structural works where electromechanical equipment is to be installed shall be inspected to ensure conformance with designs and equipment installation requirements.
- **Installation Inspections and Tests:** A system of inspections and tests, as specified in the contract or recommended by the equipment manufacturer, shall be employed throughout movement to position and installation of equipment and systems. Inspections shall be performed by PMC at critical points during installation. Surveillance shall be provided by PIU/PMC throughout the progress of work to ensure that installation is performed in accordance with the contract requirements, approved drawings, acceptable workmanship standards and configuration control requirements. All field modifications and retrofit work shall be performed under the surveillance of the PIU and PMC installation inspector.
- **Installation Verification Inspections:** Prior to all mechanical and electrical testing, verification inspections shall be performed to ensure that equipment has been satisfactorily installed.
- **System Tests:** These tests shall be conducted as appropriate to demonstrate that the installed systems are free from damage due to shipment and installation, and that equipment performs in accordance with specifications.
- **Integrated Tests:** After completion of system tests, integrated tests shall be performed to demonstrate that the system performs satisfactorily when connected to its interfacing systems or sub-systems. These tests will be followed up by commissioning tests.
- **Commissioning Tests:** These consist of a series of tests performed under service operating procedures to demonstrate compatibility of the physical plant with operating procedures.
- **Final Inspections:** Final inspections shall be performed to ensure that the completed work is in accordance with the contract and that all previously identified discrepancies have been resolved satisfactorily.

10. DOCUMENT CONTROL

Document control is intended to provide a consistent framework for transmittal, receipt, recording, processing, filing and retrieval of documents, and to ensure commonality in formats. The most important documents for QA/QC are final design documents, test reports and instructions. A flow chart for control of these documents is shown in Figure 10.1. Document control procedures, including guidelines for correspondence control, are outlined below.

10.1 Design Document Control

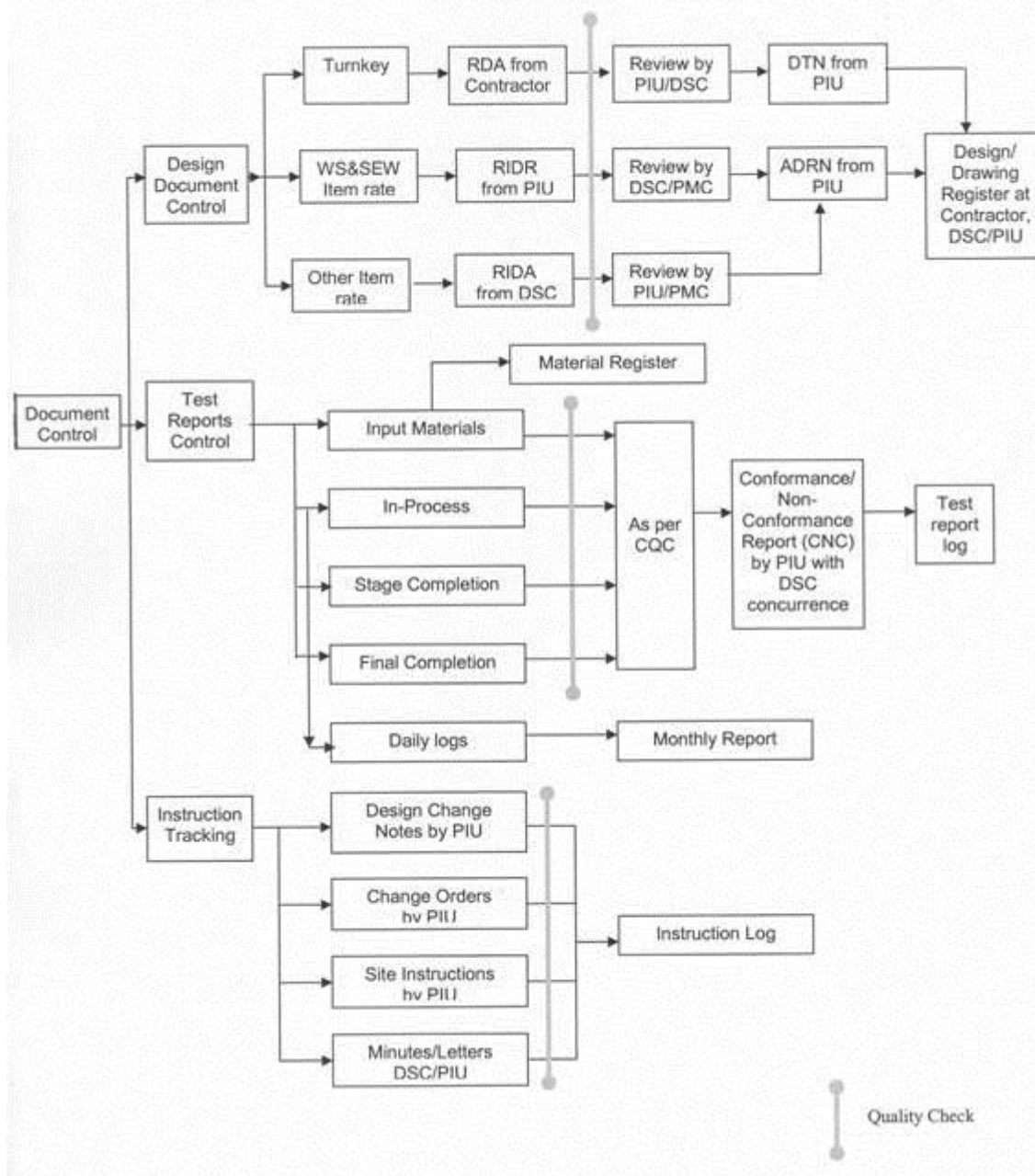
As discussed earlier (See Section 3), detail design documents (drawings, calculations, estimates, etc.) are generated at two separate levels, depending on the contracting procedure and type of work. These factors have been considered in suggesting the design document control system.

10.1.1 Turnkey contracts

The flow of detail design documents prepared by the contractor in turnkey contracts shall be as follows:

1. The contractor shall submit three copies of design documents to PIU for review, using the Request for Design Approval (RDA) Format F-1 of Appendix C.
2. The PIU shall send one copy of the documents to PMC for review and comments, using the Request for Internal Design check Review and recommendation (RIDCRR) Format F-2 of Appendix C.
3. After review, the PMC shall return the documents with its comments, using the Internal Design check Review and recommendation Note (RIDCRR) Format F-3 of Appendix C.
4. Taking into account the comments of PMC, the PIU shall get it reviewed & checked from with its recommendation for approvals & not approval using (RIDCRR) format F-2 and accordingly return the design documents to the contractor, using the Design Transmittal Note (DTN) Format F-4 of Appendix C. Design documents shall be marked (or stamped) 'Approved', 'Approved as Noted' or 'Not Approved'.
5. For design documents marked 'Not Approved', steps 1 to 4 above shall be repeated. For documents noted 'Approved' or 'Approved as Noted', the contractor shall submit originals to PIU for affixing 'Approved' signatures (using a format similar to F-1.).
6. The PIU shall have the design documents signed 'Approved' and return them to the contractor (using a format similar to F-4).
7. The design document details are to be recorded in the contractor's Design/Drawing Register, using the Format F-5 of Appendix C. (Similar registers shall also be kept by PIU and PMC.)

Figure 10.1
FLOW CHART FOR DOCUMENT CONTROL



10.1.2 Item rate contracts (Working Drawings)

Item Rate Contracts are those for Water supply, road restorations, buildings, site and services, etc. The document control procedure is similar to that for water supply and sewerage item rate contracts, except that the final design documents are prepared by the PMC. The flow of documents is summarized below.

1. Copies of design documents prepared by PMC shall be forwarded to PIU and for review, using the Request for Internal Design Approval (RIDA) Format F-8 of Appendix C with copy to PIU for information.
2. After review, shall forward to PIU the documents with its comments, using the Internal Design Review Note (IDRN) Format F-3 of Appendix C.
3. Taking into account the comments of PIU/ for modification if any , the PIU shall return the design documents to PMC, using the Internal Design Transmittal Note (IDTN) format F-9 of Appendix C. Design documents shall be marked (or stamped) 'Approved ', 'Recommended for approval as Noted' or 'Not Approved'.
4. For design documents marked 'Recommended for approval as Noted', and 'Not Approved', steps 1 to 3 above shall be repeated. For documents noted 'Approved', the PMC shall submit originals to PIU/PIU for affixing 'Approved' signatures (using a format similar to F-8.). However in special circumstances drawing marked 'Recommended for Approval as Noted' may be used for construction with special permission.
5. After the design documents have been signed 'Approved' for construction, they shall be transmitted to the contractor by PIU, using the Approved Design Release Note (ADRN) Format F-6 of Appendix C.
6. The document details are to be recorded in the contractor's Design/Drawing Register, using Format F-7 of Appendix C. (Similar registers shall also be kept by PMCs.)

10.2 Test Report Controls

All the tests and field checks are to be carried out as per the applicable quality control requirements. The flow of test report documentation shall generally be as follows:

1. Test reports shall be submitted by the contractor to the PMC.
2. The PMC shall forward a copy of the test reports to PIU for its review in its monthly report along with contractor's monthly report.
3. The PMC shall issue a Conformance/Non-Conformance Report (CNC Report) to the contractor after review of test results by the PMC, using Format F-10 of Appendix C. The CNC reports will have a running serial number for each contract package.
4. The CNC report shall be entered in the Test Report Log by the contractor at the site, using Format F-11 of Appendix C. The details of input materials will be recorded in the Material Register, using Format F-12 of Appendix C. The contractor shall maintain all test records properly.

Other approvals given to the contractor will be recorded in the daily logs of the contractor which should form part of the contractor's monthly report. A recommended format for Daily Work Record/Site Order Book is illustrated in Format F-13 of Appendix C.

Similar procedures shall be followed for the transmittal and review of test reports for tests performed at outside laboratories, for manufacturers' certificates, and for third party inspection reports.

10.3 Tracking of Instructions

During the process of construction, different agencies are expected to conduct site visits and instruct the relevant agency to ensure quality and timely construction within the costs to the extent possible. The multiplicity of agencies is a special feature of the Project. Hence there may be some ambiguity in the instruction flow if these are not transmitted and recorded properly.

All the instructions to the contractor shall flow through PMC/PIU. The instructions are of the following types:

1. All instructions related to the contract administration including approval of the contract variation orders, time extensions, notices related to rate of progress etc..
2. The instructions regarding quality, testing, monitoring and work scheduling shall be issued by the PMC.
3. Instructions issued during site visits or inspections of the PIU, , PIU and PMC, which are normally recorded in the contractor's Site Order Book, shown in Format F-13 of Appendix C; and
4. Instructions issued during review meetings in the form of minutes, letters, etc.

All instructions to the contractor noted above are to be recorded by the contractor in the Instruction Log, using Format F-15 of Appendix C. Instructions also include notices of rejection of work inspected because it was found to be non-conforming to requirements and which has to be redone or rectified.

10.4 Site Order Book

The Contractor shall be responsible to maintain a Site Order Book, in duplicate, at the site of the works at all times, and this shall be open for inspection by authorized representatives of the , the PIU and the PMC.

The Site Order Book has two primary purposes – to record the day-to-day instructions to the Contractor and the Contractor's compliance with these instructions, and to record the inspection and acceptance of work completion stages along with issuing approvals to the Contractor to proceed with the next stage of construction.

As noted above, the status of the Contractor's compliance with instructions issued is to be summarized in the Instruction Log (Format F-15 of Appendix C), and reviewed monthly by the PMC/PIU and during the periodic Squad Checks. In cases where the Contractor has failed to comply with the instructions, the reasons therefore shall be determined and necessary remedial actions taken. The Support Engineer of PMC may also maintain a parallel site order Book to ensure compliance.

10.5 Correspondence Control

Out-going letters (including transmittal letters and notes) originating from various organizations involved in the Project (PIU, PIUs, , PMCs, contractors, manufacturers, etc.) shall be signed only by the designated project executive of that organization (for example PIU Director or person authorized on his behalf, PIU Executive Engineer, Team Leader, PMC).

All letters should have a reference code and number, and should refer to a single subject only, which shall be clearly stated on top of the letter, after the recipient's address. All

outgoing letters should be numbered sequentially. All replies should refer to the originator's reference code and number and subject.

Incoming correspondence should be stamped and dated, and preferably given an internal reference code and number. All incoming and outgoing correspondence should be logged chronologically, either in computer correspondence registers or in manual correspondence logs.

Copies of outgoing correspondence and originals of incoming should be filed in chronological files at the document center of each project organization. There should be only one chronological file for all outgoing correspondence. Regarding incoming correspondence, there could be more than one chronological file, based on the volume of correspondence expected to be received from project related organizations (for example IAs can maintain separate incoming chronological files for each contractor, etc.)

10.6 Inspection of Site Documentation

During site visits and inspections, PMC and PIU will check and follow up with the documentation maintained on site by the contractor, as follows:

- Check the Design/Drawing Register and ensure that the approved designs and drawings are being used during construction.
- Check the Test Report Log and ensure that a Conformance Report has been issued by the PMC for the materials being used.
- Check the Material Register and cross-check the material test reports with inward and consumption entries of the Material Register.
- Check the Instruction Log and the Daily Work Record/Site Order Book and ensure that the instructions, as recorded in these registers and issued through any letters or minutes of meeting, are being implemented by the contractor.
- Check the Daily Logs and the standard of works and documentation is of acceptable quality.
- Deviations, if any, are to be recorded in the Site Order Book and a copy circulated by the contractor to the PIU, , PIU and PMC.
- The registers that are verified by the PIU and PMC are to be signed by them.

11. REPORTING

This section of the QA/QC Manual outlines the Project's requirements for progress reporting and suggests formats for reports.

11.1 Types of Progress Reports

The Project provides for five main levels of reporting, as follows:

1. Contractors' Monthly Progress Reports;
2. Project Management Consultant's Monthly Progress Reports; and
3. Quarterly Progress Reports

11.2 Reporting Schedule and Distribution

The timetables for submission of progress reports and distribution requirements are summarized below.

11.2.1 Contractors' progress reports

The contractor's Monthly Progress Report, along with monthly running bill, shall be submitted to the PIU by the 15th of the month (original plus 5 copies). To facilitate timely submission, joint measurements shall be taken by the Contractor and the PMC by the 15th of each month, prior to close of the reporting period. The reporting period would be 16th of previous month to 15th of the month. The PIU should pass the bill for action to PMC in one day after checking its admissibility as per the appendix to the contract.

11.2.2 Monthly contract reports

The PMC shall prepare a Monthly Progress Report for each contract, consisting of (i) a "Statement of Exceptions" commentary on the contractor's progress report, and (ii) a discussion of the major problems and actions taken or proposed to be taken. This shall be distributed, together with a copy of the Contractor's report, by the 5th of the month, to:

- PIU
- GMDA

The PMC would also submit the monthly running bill received by him from the PIU duly verified and entered as accepted in measurement book (PMC will make quality certification and 100% check of measurements) along with his monthly progress report copy to PIU by the 20th of each month.

11.2.3 Monthly finance reports

The PIU will send a monthly finance report to the GMDA by 31st of every month to report the funds receipt and utilization, statement of expenditure pending liabilities and expenditure forecasts.

The PMC shall also prepare a Monthly Financial Report for the works under his jurisdiction consisting of (i) a "Summary Statement of Fund Utilization", and (ii) a "Statement of Expenditures" report(s) in the approved JICA format and submit to PIU with supporting documentation by the 25th of each month. Copies of "Summary Statement of Fund Utilization" are to be provided to and PIU.

11.2.4 Project management consultant's progress reports

The shall prepare and issue a consolidated Monthly Progress Report for the Project as-a-whole by the 5th of the month.

11.2.5 Quarterly progress reports

Based on the monthly progress reports, PMC shall assist PIU in preparing and issuing Quarterly Progress Reports by the 20th of the month following the end of the Quarter for submittal to the JICA, Empowered Committee and all involved agencies.

11.3 Report Formats

Suggested formats for contractors' monthly reports is presented in Appendix D.

APPENDICES

APPENDIX A

TEST REPORT FORMATS

APPENDIX B

INSPECTION CHECKLISTS

APPENDIX C

DOCUMENTATION FORMATS

APPENDIX D

MONTHLY REPORT FORMATS

APPENDIX A

TEST REPORT FORMATS

Sl. No	Test Report Title	Format No.	Page No.
1.	Test Certificate for Cement	TC-M-01-01	
2.	Test Certificate for Sand	TC-M-02-01	
3.	Test Certificate for Water for Construction Works	TC-M-03-01	
4.	Test Certificate for Bricks	TC-M-04-01	
5.	Water Absorption Test	TC-M-05-01	
6.	Aggregate Impact Value Test	TC-M-06-01/1	
7.	Los Angeles Abrasion Test	TC-M-06-01/2	
8.	Soundness Test	TC-M-06/02	
9.	Flakiness and Elongation Index Test	TC-M-06-03	
10.	Gradation/Sieve Analysis	TC-M-06-04	
11.	California Bearing Ratio Test (CBR)	TC-M-07-01	
12.	Moisture Content Test (Field)	TC-M-07-02	
13.	Free Swell Index Test for Soils	TC-M-09-01	
14.	Liquid Limit and Plasticity Index (Atterberg Limits)	TC-M-09-02	
15.	OMC & MDD Test	TC-M-09-03	
16.	Grain Size Distribution	TC-M-09-04	
17.	Penetration Test for Grading Bitumen	TC-M-10-01	
18.	Ductility Test for Bitumen	TC-M-10-02	
19.	Bituminous Mix Dispatch Slip	TC-M-10-03	
20.	Stripping Value for Aggregates	TC-M-11-01	
21.	Approval of Borrow Material Source for Soil	TC-M-15-01	
22.	Concrete Compressive Strength Test	TC-G-01-01	
23.	Concrete Slump Test	TC-G-01-02	
24.	Daily Concrete Report	TC-G-01-03	
25.	Consistency of Mortar Test	TC-G-02-01	
26.	Rate of Spreading for Road Works	TC-R-02-01	
27.	Marshal Stability Test	TC-R-05-01	
28.	Field Density Test by Sand Replacement Method	TC-R-06-01	
29.	Core Test for Compacted Layer for BM/DBM/BC	TC-R-07-01	
30.	Surface Regularity and Control of Alignment	TC-R-07-02	
31.	Hydrostatic Test for NP Pipes	TC-P-04-01	
32.	Hydrostatic Test for Pressure Pipes	TC-P-04-02	
33.	Leak Test for Manholes	TC-P-05-01	
34.	Leak Test for Underground RCC Structures	TC-P-06-01	
35.	Leak Test for Elevated RCC Structures	TC-P-06-02	

TEST CERTIFICATE FOR CEMENT**Format No.:TC-M-01-01**

Lab Ref. No.: _____

Date _____

Package Name:		Package No.:	
Contractor Name:		Contract No.:	
Brand & Type of Cement :		Quantity : _____ T, _____ Bags	
Consignment No.: _____	No. of Samples Tested: _____	Date	Sample Collected : _____
Stores Entry No.: _____	Tested as per IS- _____		Tested : _____

1. CONSISTENCY

Trial No.	Wt. of cement (gms)	Wt. of water added (gms)	Percentage of water (%)	Reading of indicator (mm)	Consistency (P)	Remarks

2. SETTING TIME

Setting Time	Time recorded when water added	Time recorded at set	Setting time	Remarks
Initial Set				
Final Set				

3. FINENESS (BY DRY SIEVING)

Wt. of cement used	Retained on 90 micron IS Sieve	Percentage retained	Remarks

4. COMPRESSIVE STRENGTH (Check for 72, 168 & 672 hrs.)

Room Temp	Date of		Age of specimen	Crushing load (T)	Crushing Strength kg/cm ²	Remarks
	Casting	Testing				

(Cube Size = 7.06 cm, Wt. of Cement = 200 gms, Wt. of Standard Sand = 600 gms)

Comments of Laboratory In-Charge _____

Signed & Sealed by Laboratory In-Charge _____

Witnessed by: _____

Name

Designation

Signature

TEST CERTIFICATE FOR SAND**Format No.:TC-M-02-01**

Lab Ref. No: _____

Date _____

Package Name:		Package No.:	
Contractor Name:		Contract No.:	
Source of Supply:		Quantity : _____ Nos.	
Consignment No.:	No. of Samples Tested: _____	Date	Sample Collected: _____
Stores Entry No.:	Tested as per IS- _____		Tested : _____

Whether Bulking Test Carried out for this Source : YES/NO,

If yes, give Lab Ref. No.: _____, Date _____

Dry Bulk Density (Loose): _____

1. SIEVE ANALYSIS (IS-2386)

Sieve No.	Sieve Size (mm)	Retained on Each Sieve (% Wt)	Cumulative Percentage Retained	Passing Through (%Wt)	Grading Limits/Remarks

2. FINENESS MODULUS : _____
3. DELTERIOUS MATERIAL: Type of Material _____
 % by Weight _____

Comments of Laboratory In-Charge

Signed & Sealed by Laboratory In-Charge

Witnessed by:

Name

Designation

Signature

TEST CERTIFICATE FOR WATER FOR CONSTRUCTION WORK**Format No.:TC-M-03-01**

Lab Ref. No: _____

Date _____

Package Name:		Package No.:	
Contractor Name:		Contract No.:	
Source of Supply:			
	No. of Samples Tested: _____ Tested as per IS- _____	Date	Sample Collected: _____ Tested : _____

Sl.No	Color	pH Value	Organic Solids (mg/l)	Inorganic Solids (mg/l)	Sulphates (as SO ₄) (mg/l)	Chlorides (as Cl) (mg/l)	Suspended Matter (mg/l)
<i>Limits</i>		>6	<200	<3000	<500	PCC <2000 RCC <1000	<2000

Traces of Oil & Grease _____

Average Hardness as CaCO₃ _____
Comments of Laboratory In-Charge

Signed & Sealed by Laboratory In-Charge

Witnessed by: _____

Name
Designation
Signature

TEST CERTIFICATE FOR BRICKS**Format No.:TC-M-04-01**

Lab Ref. No: _____

Date _____

Package Name:		Package No.:	
Contractor Name:		Contract No.:	
Source of Supply:		Quantity : _____ Nos.	
Consignment No.:	No. of Samples Tested: _____	Date	Sample Collected: _____
Stores Entry No.:	Tested as per IS- _____		Tested : _____

1. PHYSICAL PROPERTIES

Dimensional Conformance	
Sounding	
Color	
Foreign Particles	
Obliqueness	

2. COMPRESSIVE STRENGTH

Sl.No	Frog Mark	Size of Brick LxBxH mm	Crushing Surface Area (cm ²)	Crushing Load (T)	Crushing Strength (kg/cm ²)	Remarks

3. WATER ABSORPTION TEST

Sl.No	Frog Mark	Wt. of Oven Dried Brick (gms)	Wt. of Wet Brick (gms)	Wt. of Water Absorbed (gms)	% of Water Absorbed	Remarks

Comments of Laboratory In-Charge

Signed & Sealed by Laboratory In-Charge

Witnessed by:

Name

Designation

Signature

WATER ABSORPTION TEST**Format No.:TC-M-05-01**

Lab Ref. No: _____

Date _____

Package Name:		Package No.:	
Contractor Name:		Contract No.:	
Source of Supply:		Quantity : _____ Nos.	
Consignment No.:	No. of Samples Tested: _____	Date	Sample Collected: _____
Stores Entry No.:	Tested as per IS- _____		Tested : _____
Material Name :			

Sl.No	Wt. of Oven Dried Material (gms)	Wt. of Wet Material (gms)	Wt. of Water Absorbed (gms)	% of Water Absorbed	Remarks

Water Absorption of Material : _____

Comments of Laboratory In-Charge

Signed & Sealed by Laboratory In-Charge

Witnessed by:

Name

Designation

Signature

AGGREGATE IMPACT VALUE TEST**Format No.:**TC-M-06-01/1

Lab Ref. No.: _____

Date _____

Package Name:		Package No.:	
Contractor Name:		Contract No.:	
Source of Supply:		Quantity : _____ Nos.	
Consignment No.:	No. of Samples Tested: _____	Date	Sample Collected: _____
Stores Entry No.:	Tested as per IS- _____		Tested : _____
Material Name :		No. of Std. Blows : 15	Height of Fall : 380 mm

No.	Detail	Unit	Trial				
			1	2	3	4	5
1	Wt. of dry aggregate passing 12.5 mm and retained on 10 mm sieve + cylinder measure	Gms					
2	Wt. of cylindrical measure	Gms					
3	Wt. of dry aggregate taken (1)-(2)	Gms					
4	Wt. of crushed aggregate passing 2.36 mm sieve after subjecting the test specimen to 15 blows	Gms					
5	Aggregate Impact Value $100 \times (4)/(3)$	%					

Average Value : _____

Comments of Laboratory In-Charge

Signed & Sealed by Laboratory In-Charge

Witnessed by:

Name

Designation

Signature

LOS ANGELES ABRASION TEST**Format No.:**TC-M-06-01/2

Lab Ref. No: _____

Date _____

Package Name:		Package No.:	
Contractor Name:		Contract No.:	
Source of Supply:		Quantity : _____ Nos.	
Consignment No.:	No. of Samples Tested:_____	Date	Sample Collected: _____
Stores Entry No.:	Tested as per IS-_____		Tested : _____
Material Name :		No. Of Revolutions 500/100 rpm	No. Of Abrasive Changes

No.	Detail	Unit	Trial				
			1	2	3	4	5
1	Weight of aggregate of specified gradation	Kg					
2	Weight of passing IS 1.7 mm sieve after specified revolution	Kg					
3	Los Angeles Abrasion Value =100*(2)/(1)	%					

Average Value : _____

Comments of Laboratory In-Charge

Signed & Sealed by Laboratory In-Charge

Witnessed by:

Name

Designation

Signature

SOUNDNESS TEST**Format No.:**TC-M-06-02

Lab Ref. No.: _____

Date: _____

Package Name:		Package No.:	
Contractor Name:		Contract No.:	
Source of Supply:		Quantity : ____ Nos.	
Consignment No.:	No. of Samples Tested: _____	Date	Sample Collected: _____
Stores Entry No.:	Tested as per IS- _____		Tested : _____
Material Name :		Immersion Time	Type of Solution

No.	Detail	Unit	Number of Cycle									
			1	2	3	4	5	6	7	8	9	10
1	Wt. Of clean dry aggregate of specified size range	Gms										
2	Wt. of dry aggregate dried in oven after each cycle	Gms										
3	Loss of aggregate after 10 number of cycles subject to immersion and oven drying after each cycle =(1)-(2)	Gms										
4	Soundness Value =100*(3) / (1)											

Average Value : _____

Comments of Laboratory In-Charge

Signed & Sealed by Laboratory In-Charge

Witnessed by:

Name

Designation

Signature

FLAKINESS AND ELONGATION INDEX TEST**Format No.:TC-M-06-03**

Lab Ref. No: _____

Date _____

Package Name:		Package No.:	
Contractor Name:		Contract No.:	
Source of Supply:		Quantity : _____ Nos.	
Consignment No.:	No. of Samples Tested: _____	Date	Sample Collected: _____
Stores Entry No.:	Tested as per IS- _____		Tested : _____
Material Name :			

Sieve Range (mm)	Total Wt. of Aggregate (gms)	Wt. Of Aggregate Passing Thickness Gauge	Flakiness Index	Wt. Of Non-flaky Sample	Wt. Of Aggregate Passing Elongation Gauge	Elongation Index	
63-50							
50-40							
40-31.5							
31.5-25							
25-20							
20-16							
16-12.5							
12.5-10							
10-6.3							

Flakiness Index _____

Elongation Index _____

 Comments of Laboratory In-Charge _____

Signed & Sealed by Laboratory In-Charge

 Witnessed by: _____

Name
Designation
Signature

GRADATION/SIEVE ANALYSIS**Format No.:TC-M-06-04**

Lab Ref. No: _____

Date _____

Package Name:		Package No.:	
Contractor Name:		Contract No.:	
Source of Supply:		Quantity : _____ Nos.	
Consignment No.:	No. of Samples Tested:_____	Date	Sample Collected: _____
Stores Entry No.:	Tested as per IS-_____		Tested : _____
Material Name :			

Sieve No.	Sieve Size (mm)	Wt. Of material retained (g)	Retained on Each Sieve (% Wt)	Cumulative Percentage Retained	Passing Through (%Wt)	Grading Limits/Remarks

Comments of Laboratory In-Charge

Signed & Sealed by Laboratory In-Charge

Witnessed by:

Name

Designation

Signature

CALIFORNIA BEARING RATIO (CBR) TEST**Format No.:**TC-M-07-01

Lab Ref. No.: _____

Date _____

Package Name:		Package No.:	
Contractor Name:		Contract No.:	
Source of Supply:		Quantity : _____ Nos.	
Consignment No.:	No. of Samples Tested: _____	Date	Sample Collected: _____
Stores Entry No.:	Tested as per IS- _____		Tested : _____
Specimen Preparation Method			
Type of Compaction Used			
Condition of Specimen at Test			

Density Determination			Moisture Content		Before Compaction			After Compaction		
Mould No.			Mould No.							
			Cup No.							
Wt. of mould+soil			Wt. Of cup+wet soil							
Wt. of mould			Wt. Of cup+dry soil							
Wt. of compacted soil			Wt. Of water							
Volume of mould			Wt. Of cup							
Bulk density (g/cc)			Wt. Of dry soil							
Dry density (g/cc)			Moisture content %							
MDD			OMC							
Degree of compaction			Variation							

Mould No.	Proving Ring Reading	Load (kg/cm ²)	Proving Ring Reading	Load (kg/cm ²)	Proving Ring Reading	Load (kg/cm ²)
Penetration (mm)						
0						
0.5						
1.0						
1.5						
2.0						
2.5						
3.0						
4.0						
5.0						
7.5						
10.0						
12.5						
Swell	Initial Reading	Final Reading	Net Swell	Swell		Average Swell
Mould No.						

CBR at 2.5 mm _____ CBR at 5.0 mm _____ Average CBR value at penetration ____ mm = %

Signed and Sealed by Laboratory In-Charge

Witnessed by:

Name

Designation

Signature

MOISTURE CONTENT TEST (FIELD)**Format No.:TC-M-07-02**

Lab Ref. No: _____

Date _____

Package Name:		Package No.:	
Contractor Name:		Contract No.:	
Source of Supply:		Quantity : _____ Nos.	
Consignment No.:	No. of Samples Tested:_____	Date	Sample Collected: _____
Stores Entry No.:	Tested as per IS-_____		Tested : _____

Test No.	1	2	3	4	5
Container No.					
Wt. Of wet soil taken (S_w)					
Wt. Of dry soil in pan (S_d)					
Wt. Of water ($W=S_w-S_d$)					
Moisture content = $100 * W/S_d$					

Comments of the Laboratory In-charge

Signed and Sealed by Laboratory In-Charge

Witnessed by:

Name

Designation

Signature

FREE SWELL INDEX TEST FOR SOILS**Format No.:TC-M-09-01**

Lab Ref. No: _____

Date _____

Package Name:		Package No.:	
Contractor Name:		Contract No.:	
Source Of Supply:		Quantity : _____ T/M ³	
Consignment No.: _____	No. of Samples Tested: _____	Date	Sample Collected : _____
Stores Entry No.: _____	Tested as per IS- _____		Tested : _____
Oven Dry Soil Passing Through 425-micron IS Sieve	Weight of Each Sample _____ gms	Glass Graduated Cylinders _____ ml size	

1. Volume of Soil Specimen read from the Graduated Cylinder
Containing Distilled Water (V_d) _____ ml
2. Volume of Soil Specimen read from the Graduated Cylinder
Containing Kerosene (V_k) _____ ml
3. Free Swell Index .. $100 \times (V_d - V_k) / V_k$ _____ %

Comments of Laboratory In-Charge

Signed & Sealed by Laboratory In-Charge

Witnessed by:

Name

Designation

Signature

LIQUID LIMIT AND PLASTICITY INDEX (Atterberg limits)**Format No.:TC-M-09-02**

Lab Ref. No: _____

Date _____

Package Name:		Package No.:	
Contractor Name:		Contract No.:	
Source Of Supply:		Quantity : _____ T/M ³	
Consignment No.: _____	No. of Samples Tested: _____	Date	Sample Collected : _____
Stores Entry No.: _____	Tested as per IS- _____		Tested : _____

Sr. No.	Particular	Unit	Liquid Limit					Plastic limit		
			1	2	3	4	5	1	2	3
1.	No. of blows	No								
2.	Cup No.									
3.	Wt. of cup+wet soil (W1)	g								
4.	Wt. of cup+dry soil (W2)	g								
5.	Wt. of water = W3=(W1-W2)	g								
6.	Wt. of cup = W4	g								
7.	Wt. of dry soil W5=W2-W4	g								
8.	Moisture content $W=W3/W5 \times 100$	%								

Liquid limit W_L : %Plastic limit, W_p : %Plasticity index, $I_p = W_L - W_p =$ %

Comments of Laboratory In-Charge

Signed & Sealed by Laboratory In-Charge

Witnessed by:

Name

Designation

Signature

OMC & MDD TEST**Format No.:TC-M-09-03**

Lab Ref. No.: _____

Date _____

Package Name:		Package No.:	
Contractor Name:		Contract No.:	
Brand & Type of Cement :		Quantity : _____ T/M ³	
Consignment No.: _____	No. of Samples Tested: _____	Date	Sample Collected : _____
Stores Entry No.: _____	Tested as per IS- _____		Tested : _____

a) Bulk density

Sr. No.	Particular	Unit	Test no.							
			1	2	3	4	5	6	7	8
1.	Wt. of mould + compacted soil (W1)									
2.	Wt. of mould (W2)	g								
3.	Wt. of compacted soil (W1 - W2)	g								
4.	Volume of mould (V)	cc								
5.	Wet density (rb) (W1-W2)/V	g/cc								

b) Dry density

Sr. No.	Particular	Unit	Test no.							
			1	2	3	4	5	6	7	8
1.	Wt. of cup + wet soil (W3)	g								
2.	Wt. of cup + dry soil (W4)	g								
3.	Wt. of water = W5=(W3-W4)	g								
4.	Wt. of cup (W6)	g								
5.	Wt. of dry soil, W7=W4-W6	g								
6.	Moisture content =W=(W5/W7)x100	%								
7.	Dry density = rd=rb(1+W/100)	g/cc								

Maximum dry density

Mean MDD : g/cc _____

Mean OMC : % _____

Comments of Laboratory In-Charge

Signed & Sealed by Laboratory In-Charge

Witnessed by:

Name

Designation

Signature

GRAIN SIZE DISTRIBUTION GRAPH**Format No.:TC-M-09-04**

Lab Ref. No.: _____

Date _____

Package Name:		Package No.:	
Contractor Name:		Contract No.:	
Source Of Supply:		Quantity : _____ T/M ³	
Consignment No.: _____	No. of Samples Tested: _____	Date	Sample Collected : _____
Stores Entry No.: _____	Tested as per IS- _____		Tested : _____

1. Description of soil : _____
2. Type of sieve analysis : day/wet
3. Total wt. of soil sample : W (g) = _____

IS Sieve opening	Wt. of sieve dish (g)	Wt. of sieve + dry soil (g)	Wt. of soil retained (g)	Cumulative wt. retained (g)	Cumulative percent retained	----
100 mm						
63 mm						
22 mm						
6.3 mm						
4.75 mm						
2.0 mm						
600 micron						
212 micron						
75 micron						
63 micron						
Passing 63 micron						

Report on gradation of curve :

Uniformity coefficient = $C_u = D_{60}/D_{10} =$ _____

Comments of Laboratory In-Charge _____

Signed & Sealed by Laboratory In-Charge

Witnessed by: _____

Name

Designation

Signature

PENETRATION TEST FOR GRADING BITUMEN**Format No.:TC-M-10-01**

Lab Ref. No: _____

Date _____

Package Name:		Package No.:	
Contractor Name:		Contract No.:	
Brand & Type of Cement :		Quantity : _____ T	
Consignment No.: _____	No. of Samples Tested: _____	Date	Sample Collected : _____
Stores Entry No.: _____	Tested as per IS- _____		Tested : _____

1. Pouring temperature, °C :
2. Period of cooling in atmosphere, minutes :
3. Room temperatures, °C :
4. Period of cooling in water bath, minutes :
5. Actual test temperatures, °C :

Sr. No.	Particulars	Test no.		
		1	2	3
1.	Initial Penetrometer dial reading (R1)			
2.	Final Penetrometer dial reading (R2)			
3.	Penetration value R2 - R1			

Mean Penetration Value

Comments of Laboratory In-Charge

Signed & Sealed by Laboratory In-Charge

Witnessed by:

Name	Designation	Signature

DUCTILITY TEST FOR BITUMEN**Format No.:TC-M-10-02**

Lab Ref. No: _____

Date _____

Package Name:		Package No.:	
Contractor Name:		Contract No.:	
Source of Supply :		Quantity : ____ T	
Consignment No.: _____	No. of Samples Tested: _____	Date	Sample Collected : _____
Stores Entry No.: _____	Tested as per IS- _____		Tested : _____

1. Grade of bitumen :
2. Pouring temperatures, °C :
3. Test temperatures, °C :
4. Periods of coding, minutes :
 - a) In air :
 - b) In water bath before trimming :
 - c) In water bath after trimming :

Sr. No.	Particulars	Briquette Number		
		1	2	3
1.	Ductility (cm)			

Mean Ductility (cm) :

Comments of Laboratory In-Charge

Signed & Sealed by Laboratory In-Charge

Witnessed by:

_____	_____	_____
Name	Designation	Signature

BITUMINOUS MIX DISPATCH SLIP**Format No.:TC-M-10-03**

Lab Ref. No: _____

Date _____

Package Name:	Package No.:
Contractor Name:	Contract No.:
Location of Hot Mix Plant :	Date : Time

1. Truck No : _____
2. Weight of Loaded Truck : _____
3. Weight of Empty Truck : _____
4. Net Weight : _____
5. Temperature of Mix : _____
6. Type of Mix : _____
7. % of Bitumen : _____

Signature of Hot Mix Plant Agent

Signature of DSC/PIU Representative

STRIPPING VALUE FOR AGGREGATES**Format No.:TC-M-11-01**

Lab Ref. No: _____

Date _____

Package Name:		Package No.:	
Contractor Name:		Contract No.:	
Source of Supply:		Quantity : _____ T/M ³	
Consignment No.: _____	No. of Samples Tested: _____	Date	Sample Collected : _____
Stores Entry No.: _____	Tested as per IS- _____		Tested : _____

Sr. No.	Particulars	Unit	Test no.		
			1	2	3
1.	Type of aggregate				
2.	Type of binder				
3.	% binder used				
4.	Total wt. of aggregate	g			
5.	Total wt. of binder				
6.	Temperature of water-bath				
7.	Stripping value				

Mean stripping value (%)

Comments of Laboratory In-Charge

Signed & Sealed by Laboratory In-Charge

Witnessed by:

Name

Designation

Signature

APPROVAL OF BORROW MATERIAL SOURCE FOR SOIL TO BE USED IN EMBANKMENT/SUBGRADE/GSB

Format No.:TC-M-15-01

Lab Ref. No: _____

Date _____

Package Name:		Package No.:	
Contractor Name:		Contract No.:	
Location of Borrow Material Source:			
Inspected By	No. of Samples Tested: _____	Date	Sample Collected : _____
Inspection Date	Tested as per IS- _____		Tested : _____

Sample No	Sand content %	Wet sieve analysis					Plasticity Index %	Proctor Density g/cc	CBR %	Compaction test	
		4.75	0.60	0.30	0.15	0.075				MDD %	OMC %

Comments of the Engineer

☐

APPROVED

☐

NOT APPROVED

Give Source Reference No.

(This ref no. is to be quoted by the contractor whenever material from this source is used)

Signature of the Engineer/Consultant

Signature of the Contractor

Witnessed by:

Name

Designation

Signature

CONCRETE COMPRESSIVE STRENGTH TEST**Format No.:TC-G-01-01**

Lab Ref. No.: _____

Date _____

Package Name:		Package No.:	
Contractor Name:		Contract No.:	
Pour Card No:	Pour Card Date:	Quantity : ____ T/M ³	
Daily Log Ref. _____	No. of Samples Tested: _____	Date	Sample Collected : _____
Date _____	Tested as per IS- _____		Tested : _____

Sr. No.	Particulars	Unit	Test no.		
			1	2	3
1.	Identification mark/Sample No.				
2.	Wt. of specimen	kg			
3.	Length of specimen	cm			
4.	Breadth of specimen	cm			
5.	Height of specimen	cm			
6.	Cross sectional area of the specimen	cm ²			
7.	Crushing load	kg			
8.	Compressive strength	kg/cm ²			

Average compressive strength of concrete = _____ kg/cm²

Comments of Laboratory In-Charge

Signed & Sealed by Laboratory In-Charge

Witnessed by:

Name

Designation

Signature

CONCRETE SLUMP TEST**Format No.:TC-G-01-02**

Lab Ref. No: _____

Date _____

Package Name:		Package No.:	
Contractor Name:		Contract No.:	
Pour Card No:	Pour Card Date:	Quantity : _____ T/M ³	
Daily Log Ref. _____ Date _____	No. of Samples Tested: _____ Tested as per IS- _____	Date	Sample Collected : _____ Tested : _____

Sr. No.	Particulars	Unit	Test no.		
			1	2	3
1.	Wt. of cement	kg			
2.	Wt. of fine aggregate	kg			
3.	Wt. of coarse aggregate	kg			
4.	Water/cement ratio				
5.	Wt. of water	kg			
6.	Slump	mm			

Average Slump of concrete = _____ mm

Comments of Laboratory In-Charge

Signed and Sealed by Laboratory In-Charge

Witnessed by:

Name

Designation

Signature

DAILY CONCRETE REPORT**Format No.:TC-G-01-03**

Pour Card No. _____

Date _____

Package Name:		Package No.:	
Contractor Name:		Contract No.:	
Where the concrete used:			
Supervised By:		Quantity : _____ T/M ³	
Daily Log Ref. _____	Mix Grade: _____	Date	Sample Collected : _____
Date _____	Time Start _____ Finish _____		Tested : _____

Concrete materials per bag of cement							
Material	Size	TC Ref	Fineness module	Specific gravity	Impact value	Water absorption	Dry batch weight
Cement							
Coarse Aggregate 1 CA1							
Coarse Aggregate 2 CA2							
Fine Aggregate							
Water							
Sieve Size mm	Gradation			% Finer			
	CA1	CA2	FA	No. of Concrete Cubes for Testing			
40				Cube No			
20				Time			
4.75				Cube No			
2.36				Time			
1.18				Cube No			
0.60				Time			
0.30							
0.15							
Slump Test						Temperature Test	
Sl.No	Time	Value	Sl.No	Time	Value	Time	Air Temperature
1			5				Concrete Temperature
2			6				
3			7				
4			8				

Signature of Contractor

Signature of Engineer

Witnessed by:

Name

Designation

Signature

CONSISTENCY OF MORTAR TEST**Format No.:TC-G-02-01**

Lab Ref. No: _____

Date _____

Package Name:		Package No.:	
Contractor Name:		Contract No.:	
Pour Card No:	Pour Card Date:	Quantity : _____ T/M ³	
Daily Log Ref. _____	No. of Samples Tested: _____	Date	Sample Collected : _____
Date _____	Tested as per IS- _____		Tested : _____

Sr. No.	Particulars	Unit	Test No.	
			1	2
1.	Wt. of cement	kg		
2.	Wt. of sand	kg		
3.	Water/cement ratio			
4.	Wt. of water	kg		
5.	Dial gauge reading before penetration	mm		
6.	Dial gauge reading after penetration	mm		
7.	Consistency of mortar	mm		

Average Consistency of mortar = _____ mm

Comments of Laboratory In-Charge

Signature of Test Lab

Witnessed by:

Name

Designation

Signature

RATE OF SPREADING

Format No.:TC-R-02-01

Lab Ref. No: _____

Date _____

Package Name: _____		Package No.: _____	
Contractor Name: _____		Contract No.: _____	
Work Type: Primer/Tack Coat/Surface Dressing/MSS/Pre-mix Carpet/BM/DBM/BC Chainage: from _____ to _____		Quantity : _____ T/M ³	
Daily Log Ref. _____ Date _____	Vehicle Speed: _____ km/h Temperature: _____ Pressure: _____	Date	Sample Collected : _____ Tested : _____

TRAY IDENTIFICATION	1	2	3	4	5
Weight of empty tray (g) W_e					
Weight of empty tray+bitumen sprayed (g) W_b					
Wt. Of bitumen sprayed on tray $W_n = W_b - W_e$					
Area of tray (m ²) A					
Rate of application $= 100 * W_n / A$					

Average rate of application = _____ kg/m²

Specification limits = _____ kg/m²

Comments of Inspector

Signature of the Engineer with Date

Signature of Contractor

Witnessed by:

_____ Name

_____ Designation

_____ Signature

MARSHAL STABILITY TEST**Format No.:TC-R-05-01**

Lab Ref. No: _____

Date _____

Package Name:		Package No.:	
Contractor Name:		Contract No.:	
Hot Mix Plant Data:		Quantity : _____ T/M ³	
Daily Log Ref. _____ Date _____	No. of Samples Tested: _____ Tested as per IS- _____	Date	Sample Collected : _____ Tested : _____

Stability and flow value determinations

Type of grading of aggregates :

Grade of bitumen :

Mixing temperature, °C

Compacting temperature, °C

Number of blows on either side =

Proving ring calibration factor =

Flow value dial, 1 division =

Sample No.	Bitumen content percent	Maximum proving ring reading	Stability value, kg		Flow dial reading	Flow value, 0.25/0.1 mm units
			Measured	Corrected		
1						
2						
3						
Average						
1						
2						
3						
Average						
1						
2						
3						
Average						
1						
2						
3						
Average						
1						
2						
3						
Average						

...continued

DENSITY AND VOID DETERMINATIONS

W1 =

G1 =

W2 =

G2 =

W3 =

G3 =

W4 =

G4 =

Sample No.	Bitumen content, percent	Height of sample, mm	Weight, g		Bulk Density, G_b	V_v	V_b	VMA	VFB
			in air	in water					
1									
2									
3									
Average									
1									
2									
3									
Average									
1									
2									
3									
Average									
1									
2									
3									
Average									
1									
2									
3									
Average									

Result

- (i) Maximum stability, kg = _____, at bitumen content, % = _____
- (ii) Maximum bulk density, g/cc = _____, at bitumen content, % = _____
- (iii) Percent air voids = _____, at bitumen content, % = _____

Average bitumen content =

Comments of Laboratory In-Charge

Signed & Sealed by Laboratory In-Charge

Witnessed by:

Name

Designation

Signature

FIELD DENSITY TEST BY SAND REPLACEMENT METHOD**Format No.:TC-R-06-01**

Lab Ref. No: _____

Date _____

Package Name:		Package No.:	
Contractor Name:		Contract No.:	
		Quantity : _____ T/M ³	
Daily Log Ref. _____	No. of Samples Tested: _____	Date	Sample Collected : _____
Date _____	Tested as per IS- _____		Tested : _____

I. Determination of Bulk Density of Sand

Sl.No	Particulars	Unit	Trial No		
			1	2	3
1	Inside dia. Of calibrating container	Cm			
2	Inside height of calibrating container	Cm			
3	Volume of calibrating container	CC			
4	Wt. of sand + cylinder before pouring	Gm			
5	Wt. Of sand in the cone	Gm			
6	Wt. Of sand + cylinder after pouring in calibrating container	Gm			
7	Wt. Of sand filling with calibrating container	Gm			
8	Bulk density of sand	Gm/cc			

II. Determination of Bulk Density of Soil In-situ

Sl.No	Particulars	Unit	Trial No		
			1	2	3
1	Wt. Of wet soil from hole	Gm			
2	Wt. Of sand + cylinder after pouring into the hole	Gm			
3	Wt. Of sand in the soil	Gm			
4	Volume of the hole	CC			
5	Bulk density of soil in-situ	Gm/cc			

III. Determination of Dry Density of Soil In-situ

Sl.No	Particulars	Unit	Trial No		
			1	2	3
1	Container No.				
2	Wt. Of container	Gm			
3	Wt. Of container + wet soil	Gm			
4	Wt. Of container + dry soil	Gm			
5	Wt. Of dry soil	Gm			
6	Wt. Of water	Gm			
7	Water content %	%			
8	Dry density	Gm/cc			

Signed by Lab in-charge

Witnessed by:

Name

Designation

Signature

CORE TEST FOR COMPACTED LAYER FOR BM/DBM/BC**Format No.:TC-R-07-01**

Lab Ref. No: _____

Date _____

Package Name:		Package No.:	
Contractor Name:		Contract No.:	
Hot Mix Plant Data:		Quantity : _____ T/M ³	
Daily Log Ref. _____ Date _____	No. of Samples Tested: _____ Tested as per IS- _____	Date	Sample Collected : _____ Tested : _____

Stability and flow value determinations

Type of grading of aggregates :

Grade of bitumen :

Mixing temperature, °C

Compacting temperature, °C

Number of blows on either side =

Proving ring calibration factor =

Flow value dial, 1 division =

Sample No.	Bitumen content from analysis %	Maximum proving ring reading	Stability value, kg		Flow dial reading	Flow value, 0.25/0.1 mm units
			Measured	Corrected		
1						
2						
3						
Average						
1						
2						
3						
Average						
1						
2						
3						
Average						
1						
2						
3						
Average						
1						
2						
3						
Average						

(continued)

DENSITY AND VOID DETERMINATIONS

W1 =

G1 =

W2 =

G2 =

W3 =

G3 =

W4 =

G4 =

Sample No.	Bitumen content from analysis, %	Height of sample, mm	Weight, g		Bulk Density, G_b	V_v	V_b	VMA	VFB
			in air	in water					
1									
2									
3									
Average									
1									
2									
3									
Average									
1									
2									
3									
Average									
1									
2									
3									
Average									
1									
2									
3									
Average									

Result

- (i) Maximum stability, kg = _____, at bitumen content, % = _____
- (ii) Maximum bulk density, g/cc = _____, at bitumen content, % = _____
- (iii) Percent air voids = _____, at bitumen content, % = _____

Average bitumen content =

Comments of Laboratory In-Charge

Signed & Sealed by Laboratory In-Charge

Witnessed by:

Name

Designation

Signature

SURFACE REGULARITY AND CONTROL OF ALIGNMENT**Format No.:**TC-R-07-02**Lab Ref. No.:**_____**Date**_____

Package Name:	Package No.:
Contractor Name:	Contract No.:
Road Name:	Total Stretch Length Inspected:
Daily Log Ref:	Date Inspected
Date	Inspected By

1. Horizontal Alignment*(permissible limits as per Section 4)*

Chainage (m)		Horizontal Alignment Tolerances reckoned from center line of carriageway	
From	To	At the edges of the carriageway	At the edge of the road and lower layers

2. Tolerances in Surface Levels*(permissible limits as per Section 4)*

Chainage (m)		Sub-grade	Sub-base	Base		Wearing Course	
From	To			Machine Laid	Manual Laid	Machine Laid	Manual Laid

3. Surface Regularity of Pavement Courses*(permissible limits as per Section 4)*

Chainage (m)		Irregularity – 4 mm		Irregularity – 7 mm	
From	To	Length		Length	
		300 m	75 m	300 m	75 m
		No. of Irregularities		No. of Irregularities	

Signature of Inspecting Authority

HYDROSTATIC TEST FOR NP-PIPES**Format No.:TC-P-04-01**

Lab Ref. No: _____

Date _____

Package Name:		Package No.:	
Contractor Name:		Contract No.:	
Daily Log Ref. _____ Date _____	Tested as per CPHEEO _____	Date Tested	Test Head : 2.5 m at highest point Observation: 1 hr

Sl. No	ID (mm)	Time in hrs	Stretch	Length (m)	Water level Drop (mm)		Volume of water to restore to original level (liters)	
					10 min	30 min	10 min	30 min

Comments of the Engineer/Consultant

Any rectification to be done by the contractor

Signature of the Engineer/Consultant

Signature of the Contractor

Witnessed by:

Name

Designation

Signature

HYDROSTATIC TEST FOR PRESSURE PIPES**Format No.:TC-P-04-02**

Lab Ref. No.: _____

Date _____

Package Name:		Package No.:
Contractor Name:		Contract No.:
Daily Log Ref. _____ Date _____	Tested as per CPHEEO _____	Date Tested

Sl. No	Material of pipe	ID (mm)	Stretch	Length (m)	Applied test pressure (kg/cm ²)	Time in hrs)			Observations
						1	2	4	

Comments of the Engineer/Consultant

Any rectification to be done by the contractor

Signature of the Engineer/Consultant

Signature of the Contractor

Witnessed by:

Name

Designation

Signature

LEAK TEST FOR MANHOLE**Format No.:TC-P-05-01**

Lab Ref. No: _____

Date _____

Package Name:		Package No.:	
Contractor Name:		Contract No.:	
Daily Log Ref. _____ Date _____	Tested as per CPHEEO _____	Date Tested	Duration of Water Retention: 24 hrs

Sl. No	Manhole No.	Water Level (m)		Sign	Drop in water level (mm)	Remarks
		Initial	Final			

Comments of the Engineer/Consultant

Any rectification to be done by the contractor

Signature of the Engineer/Consultant

Signature of the Contractor

Witnessed by:

Name

Designation

Signature

LEAK TEST FOR UNDERGROUND RCC STRUCTURES**Format No.:TC-P-06-01**

Lab Ref. No: _____

Date _____

Package Name:	Package No.:
Contractor Name:	Contract No.:
Name of Structure:	Size:

Date of Filling		Initial Water level (m)		
Observation No.	Date of Observation	Water Level (m)	Drop (mm)	Remark
1				
2				
3				
4				
5				
6				
7				

Comments of the Engineer/Consultant

Any rectification to be done by the contractor

Signature of the Engineer/Consultant

Signature of the Contractor

Witnessed by:

Name

Designation

Signature

LEAK TEST FOR ELEVATED RCC STRUCTURES**Format No.:TC-P-06-02**

Lab Ref. No.: _____

Date _____

Package Name:	Package No.:
Contractor Name:	Contract No.:
Name of Structure:	Size:

Date of Filling		Initial Water level (m)	
Observation No.	Date of Observation	Observation	Remark
1			
2			
3			
4			
5			
6			
7			

Comments of the Engineer/Consultant

Any rectification to be done by the contractor

Signature of the Engineer/Consultant

Signature of the Contractor

Witnessed by:

_____	_____	_____
Name	Designation	Signature

APPENDIX B

INSPECTION CHECKLISTS

APPENDIX B.1

INSPECTION CHECKLIST FOR CONCRETING WORKS

APPENDIX B.2

INSPECTION CHECKLIST FOR STONE MASONRY WORK

APPENDIX B.3

INSPECTION CHECKLIST FOR BRICKWORK AND FINISHES

APPENDIX B.4

INSPECTION CHECKLIST FOR BUILDING SERVICES AND FINISHES

APPENDIX B.5

INSPECTION CHECKLIST FOR ROAD WORKS

APPENDIX B.6

INSPECTION CHECKLIST FOR PIPELINE WORKS

APPENDIX B.7

INSPECTION CHECK LIST FOR DRAINAGE WORK

APPENDIX B.1

INSPECTION CHECKLIST FOR CONCRETING WORKS

Items	Remarks
<u>Proportioning, Batching and Mixing of Concrete</u> ❑ The materials and proportion of concrete materials as established by the preliminary test for mix design should be rigidly followed. ❑ Concrete should be produced only by weigh batching the aggregates. The accuracy of the weigh batcher should be periodically checked and the needle should be adjusted to zero when the hopper is empty. Only a mechanical mixer machine with a hopper arrangement should be allowed for mixing concrete. ❑ The quantity of water actually entering the mixing drum should be checked with the reading of the gauge or valve setting, when starting a job. The batch should be charged into the mixer so that some water enters the drum in advance of cement and aggregate. All water should be in the drum by end of the first 15 seconds of the specified mixing time. Each batch would be mixed until the concrete is uniform in color at least for a period of two minutes after all the materials and water are in the drum. <u>Placing of Concrete</u> ❑ Before any concrete is placed the entire placing program consisting of equipment, layout, number of labors engaged, proposed procedures and methods should be submitted to the Engineer for approval. The Engineer must satisfy himself that the arrangement for mixing and conveying concrete will ensure a nearly continuous flow of concrete during depositing, contractor's labor force won't be overworked affecting quality of work, sufficient overlap is allowed between concrete placing crews working at subsequent shifts, and sufficient lighting arrangements are made if concreting is allowed in the night. ❑ The Engineer must also satisfy himself that the concrete mix design is based on truly representative samples of aggregate and cement which will be actually used during concreting. While taking representative samples from different stacks of aggregates for checking of gradation and other tests a scoop or shovel with sides should be used so that the larger stones won't roll off the side. ❑ Before the concrete is actually placed in position, the inside of the formwork should be inspected to see that they have been cleaned and oiled. All reinforcement and other items to be cast in concrete should have clean surfaces. All reinforcement should have sufficient cover, as specified in the contract, which should be ensured by using proper size cement mortar spacer blocks. The reinforcement fabrication should be finally checked ensuring proper length and spacing between reinforcement bars, proper orientation (verticality/ horizontality) of reinforcement bars and stirrups and adequate provision of development length.	

- | | |
|--|--|
| <ul style="list-style-type: none"> ❑ Concrete must be placed in its final position before the cement reaches its initial set and concrete should normally be compacted in its final position within 30 minutes of leaving the mixer. ❑ Concrete should be deposited as nearly as practicable directly in its final position and should not be re handled or caused to flow in a manner which would cause segregation, loss of material or displacement of reinforcement. Concrete should be placed in the shuttering by shovels or other approved implements and should not be dropped for a free fall from a height more than 1.0 meter. When it is necessary to use transfer chutes the Engineer must satisfy himself in respect of type, length, slopes, baffles, vertical terminals and timing of operations. To allow for loss of mortar against the sides of the chute, the first mixes should have less coarse aggregates. Concrete should not be permitted to fall free from the end of the chute by more than 1.0m and chutes should have slopes not flatter than 1 vertical : 3 horizontal but not steeper than 1 vertical : 2 horizontal. ❑ Concrete once started, should be continuous until the pour is completed and would be placed in successive horizontal layers of uniform thickness preferably ranging from 15 to 90 cm. Concrete should be placed as rapidly as practicable to prevent the formation of cold joints between each succeeding layer. The top surface of each pour and the bedding layer should be approximately horizontal. ❑ Concrete should be compacted during placing with approved vibrating equipment and consolidated to maximum practicable density, free of pockets of coarse aggregate, should fit tightly against all form surfaces, reinforcements and embedded fixtures. Caution should be exercised not to over vibrate the concrete to the point of segregation. Tapping or external vibration of forms by hand tool or immersion vibrator must not be permitted and care should be taken to prevent contact of immersion vibrators with reinforcing steel or forms of finished surfaces. ❑ Concrete should be placed without interruption until completion of part of the work between predetermined construction joints. If stopping of concrete becomes unavoidable anywhere, a properly formed construction joint shall be made where the work is stopped. Joints should be either vertical or horizontal. For inclined or curved member the joint should be at right angles to the axis of the member. The construction joint surface should be roughened, thoroughly cleaned and treated with a thin layer of cement grout before resumption of concreting. Except when placing with slip forms, each placement of concrete in multiple lift work, should be allowed to set for at least 24 hours before the start of a subsequent placement. | |
|--|--|

Curing & Protection of Concrete

- ❑ During Curing of concrete preference should be given to use of continuous spray, or ponded water, continuously saturated covering of sacking, canvas, hessian or other absorbent material, or approved effective curing compounds applied with spraying equipment capable of producing a smooth even-textured coat.
- ❑ Fresh concrete should be kept continuously wet for a minimum period of 10 days from the date of placing of concrete, following a lapse of 12-14 hours after laying concrete. However, curing of horizontal surface exposed to dry wind should begin immediately after the concrete has hardened. Water should be applied to formed surfaces immediately upon removal of forms.
- ❑ Immediately after the shuttering is removed, the Engineer should very carefully go over the entire surface of the concrete. Holes left by form bolts etc. should be filled up and made good with mortar composed of one part of cement and one and half parts of sand passing 2.36 mm IS sieve. Superficially honeycombed surface and rough patches should be similarly made good immediately after removal of shuttering and superficial water and air holes should be filled in. The mortar should be well worked into the surface with a wooden float and the surface irregularities should be removed by grinding.
- ❑ If reinforcement is exposed or honey combing occurs at vulnerable positions e.g. ends of beams or columns it may be necessary to cut the member partially or completely and reconstruct. The Engineer's decision shall be final in this regard. If only patching is necessary, the defective concrete should be cut till solid concrete is reached (or to a minimum depth of 25 mm) the edges being cut perpendicular to the affected surface or with a small undercut if possible. Anchors, tees or dovetail slots should also be provided wherever necessary to attach the new concrete securely in place.

APPENDIX B.2

INSPECTION CHECKLIST FOR STONE MASONRY WORK

Items	Remarks
❑ The type of masonry used for structural works should be random masonry (coursed or uncoursed) or coursed rubble masonry (first sort). For Bridge Works generally coursed rubble masonry should be used. ❑ Stones should be sufficiently wetted before laying to prevent absorption of water from mortar. ❑ The courses of masonry should generally be of same height. When there is to be variation in the height of courses, the larger courses are to be placed at lower levels, height of courses gradually decreasing to the top. The practice of placing loose mortar on the course and then pouring water from top must not be allowed. ❑ The stones used for hearting or interior fillings of the wall face should be laid on their broadest face that gives a better opportunity to fill the spaces between the joints. ❑ The bed which is to receive the stone should be cleaned, wetted and covered with a layer of fresh mortar. All stones should be laid full in mortar and in vertical joints. Clean chips and spalls should be wedged into the mortar joints and bed wherever necessary to avoid thick beds or joints of mortar. The face joints should not be more than 10mm and 20mm for coursed rubble and random rubble respectively. ❑ All vertical joints should be truly vertical. Vertical joints should be staggered as far as possible. Distance between the nearer vertical joint between the upper layer and lower layer should not be less than half the height of the course. ❑ Bond stones should be rectangular in shape and should be a through stone extending from face to back of the wall. For sharp corners, especially in skew bridges, through stones should also be used to avoid spalling of corners. The quoins i.e. stones specially selected and neatly dressed for forming an external angle in masonry work should not be less than as specified in the contract.	

APPENDIX B.3

INSPECTION CHECKLIST FOR BRICKWORK AND FINISHES

Items	Remarks
❑ A representative sample of bricks to be sent for testing of compressive strength, water absorption etc. should not be less than twenty bricks. These bricks must be selected at random from the particular batch. For example, if there are ten stacks of bricks, two bricks should be chosen from each stack and these must be taken from different positions in each stack. ❑ All Bricks should be thoroughly soaked in a tank filled with water for a minimum period of one hour (preferably for 24 hours) prior to being laid. However, at the time of laying, bricks should be skin dry and clean. ❑ Mortar should be mixed in the specified proportion only in such quantity as required for immediate use. Initial setting of mortar with Ordinary Portland Cement is normally considered to have taken place in 30 minutes and mortar unused for more than 30 minutes shall be rejected and removed from site of work. ❑ All Brickwork should be laid in English Bond and bricks should be laid with frogs up on a full bed of mortar. Thickness of mortar joints should not exceed 10 mm. ❑ Before laying bricks in foundation, the foundation slab should be thoroughly hacked, swept clean and wetted and a mortar layer not less than 12 mm thick should be spread on the surface of the foundation slab. ❑ When fresh masonry is to join partially/entirely set old masonry, the exposed / jointing surface of the set masonry should be cleaned, roughened and wetted to effect the best possible bonding. ❑ Brickwork should be cured keeping constantly moist on all faces for a minimum period of seven days and green work should be protected from rain by suitable covering. ❑ The surface of brickwork, which is to be subsequently plastered or painted, should be squarely raked out to a depth of 15 mm when the mortar is still green. ❑ Plastering should be started from top and worked down. The plastering surface should be finished off with a plasterer's wooden float. Metal floats shall not be used. ❑ Curing for finishes should be commenced as soon as the mortar used for finishing has hardened sufficiently, not to be damaged during curing. The finishes should be kept moist for a period of at least 7 days.	

APPENDIX B.4

INSPECTION CHECKLIST FOR BUILDING SERVICES AND FINISHES

Items	Remarks
❑ Building services and finishes work shall be closely coordinated and be undertaken in the appropriate sequence. ❑ Building service requirements including water supply pipes and fittings, sanitary fittings and appliances, electrical installation, doors and windows and ventilators etc. shall be inspected for conformance with applicable codes and specifications. ❑ All brickwork in the building will be finished with plastering whereas all stone masonry work will be finished with pointing, if not otherwise specified in the contract. Where plastering work is required, as part of the finishing work, all service lines, conduits, pipes, clamps, door and window frames and all required inserts must be in position before the plastering begins. The plastering shall be always started from the top and worked down. For purpose of giving a smooth finish to the plastering work a plasterer's wooden float should be used and no metal float should be allowed to use. Where pointing is required, the pointing will be done to give the surface an acceptable architectural look. For pointing work, the joints will be raked to a depth of 40 mm and then mortar will be placed in the joints up to a depth of 25 mm by using a trowel. ❑ Painting must be carried out in conformance with specifications. The painting materials and surface preparation must be checked prior to painting. ❑ For all finishes e.g. plastering, pointing, painting or whitewashing the contractor will first prepare a sample area and on approval of the quality of finish in the sample area, the contractor will be allowed to apply all over the building. ❑ Flooring and roofing work shall be undertaken as required by the specifications.	

APPENDIX B.5

INSPECTION CHECKLIST FOR ROAD WORKS

Items	Remarks
❑ The Engineer must check all layout and staking completed by the contractor before starting roadwork and must confirm checking at regular intervals and at completion of the work to ensure that all work conforms to plans and specifications. ❑ A careful and thorough check must be made of all subgrade areas where road materials are to be placed. No road material should be allowed to be placed unless the Engineer has visibly checked and approved the areas and has verified the quality of subgrade preparation by testing procedures as provided in the contract document and elsewhere in this document. ❑ If a question arises as to the suitability of any of the materials in meeting the specifications, the Contractor shall be responsible to provide further field and laboratory testing as may be required and requested by the field engineer. ❑ The base on which bituminous layer is to be laid, should be thoroughly swept clean, free from dust and foreign matter using mechanical broom and dust removed or blown off by compressed air. A priming coat, wherever needed, followed by a layer of tack coat shall be applied on the base before receiving the bituminous layer. A properly calibrated Bitumen Distributor Tanker having sprayer with self heating arrangement should be used for laying tack coat / prime coat. If required, the Engineer may ask for test trials to ensure proper calibration of Bitumen Distributor Tankers in order to arrive at the correct speed of vehicles, spraying temperatures etc. ❑ The Engineer must inspect Contractor's Hot Mix Plant, Bitumen Distributor Tanker / Sprayer, Tipper Trucks, Paver Finisher, Rollers and other tools and plants to be employed for bituminous work and satisfy himself with the capacity of the plants and arrangements made for mixing, laying and rolling of bituminous courses. The Engineer should receive the Job mix formula for BM/DBM/AC at least 20 days before the start of the work and must satisfy himself that the job mix formula is based on a correct and truly representative sample of the materials that will be actually used in the works. ❑ For bituminous construction, the temperature of binder at the time of mixing shall be in the range of 150°C – 163°C and that of the aggregate shall be in the range of 155°C-163°C. The difference in temperature between the binder and the aggregate in no time shall exceed 14°C.	

- | | |
|---|--|
| <ul style="list-style-type: none">❑ For bituminous construction, mixing of aggregate and binder should be thorough to ensure that a homogeneous mixture is obtained in which all particles of the aggregate are coated thoroughly and uniformly, and the discharge temperature of the mix should be in the range of 130°C-160°C. The mix should be visually inspected at the plant after delivery and transported to site by covered tipper trucks.❑ For bituminous construction, the temperature of the mix at the time of laying shall be in the range of 120°C-160°C and the temperature should be continuously monitored using a digital thermocouple based thermometer. Mixes with a temperature of less than 120°C shall not to be put for laying under any circumstances. While laying with Paver Finishers, the longitudinal joints and edges shall be constructed true to the delineating lines parallel to the centerline of the road. Longitudinal and transverse joints should be delineated at least by 250 mm from those in the lower courses and the joint in the topmost layer should not be allowed to fall within the wheel path.❑ For bituminous construction, after spreading the mix by Paver, the mix should be thoroughly compacted by rolling with a set of rollers moving at a speed not more than 5 Km/hour immediately following the Paver. For BM/ DBM / AC, the initial rolling shall be done with 8-10 T Static Weight Smooth Wheeled Roller, intermediate rolling shall be done with 8-10 T Static Weight Vibratory Roller or 15-25 T Pneumatic Tyred Roller and the finish rolling shall be done with 6-8 T Smooth Wheeled Tandem Roller. For Mix Seal Surfacing layer, rolling should be done with 8-10T Smooth Wheeled Tandem Roller. The exact pattern of rolling for all types of surfaces should be established only after a trial compaction run. The rolling should be completed in all respects before the temperature of the mix falls below 100°C. The roller should not be permitted to stand on pavement which has not been fully compacted or where the temperature of pavement is still more than 70°C.❑ Traffic on bituminous layer may only be allowed after completion of the final rolling when the mix cools down to surrounding temperature. | |
|---|--|

APPENDIX B.6

INSPECTION CHECKLIST FOR PIPELINE WORKS

Items	Remarks
❑ The engineer should ensure that all required specials, valves and pipes with jointing material duly tested are available before planning for excavation of trenches. No gaps are to be permitted with laying pipes. ❑ The engineer should also review the proposed alignment and grades and site. The pipe line should use appropriate special whenever any change in alignment and grade is required. ❑ The Engineer must ensure that pipe bedding has been prepared to the proper shape as shown in the drawing and properly compacted. Especially for laying of pipes larger than 1200 mm diameter, in earth and moorum foundation, the curvature of the bottom of the trench should match the curvature of the pipe as far as possible, subtending an angle of 120 degrees at the center of the pipe. ❑ Wherever a socket or collar of pipe or fitting/ special will occur a grip should be cut at the bottom of the trench or concrete bed to a depth of at least 75 mm below the bed of the pipe. When welding is to be carried out with the pipes and the specials in the trench, additional excavation of not more than 60 cm in depth and 90 cm in length should be made at joints in order to facilitate welding. ❑ The trench should be kept clean and dry before lowering the pipes in the trench. If subsoil water is met with during excavation contractor would provide necessary equipment and laborers for dewatering the trenches. ❑ Over excavation in regard to depth and top width of trench should not be allowed in any case. If the depth of trench is over excavated than required, contractor at his own cost will refill with gravel up to the required depth and properly compact it. Additional width, if required, would be provided only in the top portion of the ground level up to 300 mm above the crown of the pipe. If any extra width is provided in the area below this portion, the contractor shall have to provide lime concrete or rubble masonry or otherwise to the discretion of engineer. ❑ The pipes should be thoroughly investigated for any defects before lowering in the trench. The socket or spigot end of pipes should never be allowed to be chipped and any deviation in the dimension of the pipe should be reported by the contractor. The pipes should be lowered in the trench with belt slings and special care should be taken so that no metal bears against the pipe and the bearing is uniform. ❑ During jointing of pipe special care should be taken so that spigot and socket of adjoining pipes fit snugly into each other. For RCC pressure pipes, the dimension of the splayed rubber ring sitting on the groove and the clear distance available at the joint location should be checked with a gauge to ensure proper jointing of pipe.	

- | | |
|--|--|
| <ul style="list-style-type: none">❑ For water distribution network projects, water pipes, fittings and connections must be carefully inspected during placement and verified as to compliance with the plans and specifications.❑ All open ends of the pipes must be covered with suitable cover at the end of the day before close of work to prevent entry of dirt and water in the pipe.❑ Proper care to cover up the pipes satisfactorily and anchor them during laying should be taken, so as to prevent uplifting due to any entry of water in the trench. This is especially important during monsoons, which laying pipes in saturated soils and where any surface water may enter the trench.❑ Connection to existing water supply mains must be scheduled under provision of the specifications and the field engineer must actively participate in the arrangements and in insuring that the contractor perform the work as scheduled.❑ Testing of water system must be carried out in the presence of the field engineer and the entire testing procedure must be carefully checked including performing leakage testing which must meet standards as included in the specification.❑ The alignment and grade of all pipe, especially the gravity flow sewer pipes should be thoroughly checked by the Engineer. The Engineer must ensure that the pipes have been laid as per the design gradient and should also ensure that the minimum gradient has been achieved in all sections of sewer line.❑ The trenches may be back filled after pipe laying is completed, but the joints must be kept open. During back filling special care should be taken to ensure that earth lying below the pipe gets properly compacted by use of special hand tools fabricated with angled iron. The pipe main should be constructed with the earth cover as specified in the profile drawings. However, the earth cover should not be less than 1.0 m from the top of the pipe except at the locations of drainage and pipe crossings.❑ The pipe section completed and tested are best preserved by keeping them full of water till commissioning. | |
|--|--|

APPENDIX B.7

INSPECTION CHECK LIST FOR DRAINAGE WORKS

Items	Remarks
☐ Availability of equipment and machinery at site ☐ Transfer of assets from PHE and it's taking over location of services ☐ Ensure proper alignment, depth, gradient, bedding, compaction etc. ☐ Joints of pipes, pipe size, pipe type, type of bedding, saddles (chair) arrangements ☐ Locations and size of Manholes, Gully pits, connecting sewers ☐ Excavation work, shape, size of trenches, safety measures, refilling and reinstatement of road ☐ Crossing of roads, lanes, existing pipe line/sewer mains, electric and telephone cables ☐ Arrangements for traffic diversion, provision of access ☐ Construction of manholes, cu-nets, foot steps, gully pits, connecting chambers, ☐ Hydraulic testing of sewer pipes, permissible leakages from joint and remedial measures in case of leakages beyond permissible limits or any other tests called for in the technical specifications ☐ Testing of cement, steel, sand, other site tests of concrete/masonry work being carried out, results are maintained ☐ As constructed drawings and manhole, house connection records completed.	

APPENDIX C

DOCUMENTATION FORMATS

Format No.	Document Type	Responsibility
F-1	Request For Design Approval (for Turnkey Contracts)	Issued by Contractor
F-2	Request for Internal Design Check , Review and Recommendations	Issued by Implementing Agency (IA)
F-3	Internal Design check Review and Recommendation Note	Issued by PMC
F-4	Design Transmittal Note (for Turnkey Contracts)	Issued by IA
F-5	Design/Drawing Register (for Turnkey Contracts)	Maintained by Contractor
F-6	Approved Design Release Note (for All Item Rate Contracts)	Issued by IA
F-7	Design/Drawing Register (for Item Rate Contracts)	Maintained by Contractor
F-8	Request for Internal Design Approval	Issued by PMC
F-9	Internal Design Transmittal Note	Issued by IA
F-9 (a)	Design Approval Note	Issued by IA
F-10	Conformance/Non-Conformance Report	Issued by IA
F-11	Test Report Log	Maintained by Contractor
F-12	Material Register	Maintained by Contractor
F-13	Daily Work Record/Site Order Book	Maintained by Contractor
F-14	Variation Order	Issued by IA
F-15	Instruction Log	Maintained by Contractor
F-16	Minutes of Progress Review Meeting	Prepared by PMC
F-17	Interim Evaluation of Contractor Performance	Prepared by IA
F-18	Quality Certification and Acceptance	Issued by PMC

Format F-1**REQUEST FOR DESIGN APPROVAL (RDA)**

Ref: RDA_____

Date: _____

To
[PIU]

Package No. : _____ Contract No.: _____

Title of work : _____

We are herewith enclosing the following designs/drawings for your review and approval.

Design/Drawing No.	Rev	Date	Description

[Contractor]**ACKNOWLEDGEMENT COPY**

Date received _____

Ref No.: RDA_____

Date: _____

To
[Contractor]

Package No. : _____ Contract No.: _____

Title of work : _____

Received the above referred RDA along with enclosures.

[PIU]

Format F-2**REQUEST FOR INTERNAL DESIGN CHECK/REVIEW AND RECOMMENDATIONS
(RIDCRR)**

Ref:

Date: _____

To

[DSC/PMC/PIU (as applicable)]

Package No. : _____ Contract No.: _____

Title of works : _____

Please check/review and recommend on the enclosed designs/drawings and communicate your comments.

Design/Drawing No.	Rev	Date	Description

[PIU]/ DSC (as applicable)

Copy to PIU for information

Format F-3**REQUEST FOR INTERNAL DESIGN CHECK/REVIEW AND RECOMMENDATIONS
(IDCRRN)**

Ref:

Date: _____

To

[PIU]/(as applicable)

Package No. : _____ Contract No.: _____

Title of works : _____

Please find enclosed our comments and recommendations on the following designs/drawings after checking/reviewing

Design/Drawing No.	Rev	Date	Description	Comments and Recommendations

It is certified that the document has been checked reviewed and is recommended for its approval/modification.

[DSC or PMC as applicable]

Format F-4**DESIGN TRANSMITTAL NOTE (DTN)**

Ref: RDA____ DTN____

Date: _____

To
[Contractor]

Package No. : _____ Contract No.: _____

Title of work : _____

Your RDA No.: _____ Date _____

Following are our comments on the designs and drawings submitted by you under above referred RDA.

Design/Drawing No.	Rev.	Date	Description	Comments
				Not Approved, refer drawing for comments
				<i>Approved as Noted</i>
				<i>Approved</i>

[PIU]

ACKNOWLEDGEMENT COPY

Date received _____

Ref: DTN____

Date _____

To
[PIU]

Package No. : _____ Contract No.: _____

Title of work : _____

Received the above referred DTN along with enclosures.

[Contractor]

DESIGN/DRAWING REGISTER

Package No : _____
Contract No. : _____
Contractor : _____

Title of work : _____
Date started : _____
Date to complete: _____

[illegible]

Format F-6**APPROVED DESIGN RELEASE NOTE (ADRN)**

Ref: ADRN_____

Date: _____

To
[Contractor]

Package No. : _____ Contract No.: _____

Title of works : _____

Following are the approved drawings for commencing the construction. Please bring to our notice if any changes are required to be incorporated additionally.

Design/Drawing No.	Rev	Date	Description

[PIU]**ACKNOWLEDGEMENT COPY**

Date received _____

To
[PIU]

Ref: ADRN_____

Date: _____

Package No. : _____ Contract No.: _____

Title of work : _____

Received your above referred ADRN along with enclosures.

[Contractor]

Design/Drawing Register

Title of Work:

Dated Started:

[illegible]

Format F-8**REQUEST FOR INTERNAL DESIGN APPROVAL (RIDA)**

Ref:

Date: _____

To
[PIU/PMC]

Package No. : _____ Contract No.: _____

Title of works : _____

Please check/review and recommend on the enclosed modified designs/drawings and communicate your approval & signatures.

Design/Drawing No.	Rev	Date	Description

[DSC]

Certified that the designs/drawings has been reviewed and comments has been incorporated by DSC if any, and signed.

Certified that the designs/drawings has been checked, reviewed and recommended for its approval and signed.

[PMC]

Format F-9**INTERNAL DESIGN TRANSMITTAL NOTE (IDTN)**

Ref:

Date: _____

To
[DSC]

Package No. : _____ Contract No.: _____

Title of works : _____

Following are the comments from PIU/PMC on the enclosed designs/drawings submitted by you.
Submit after necessary modifications if any, for their approval.

Design/Drawing No.	Rev	Date	Description	Comments
				Not Approved, refer drawing for comments
				Recommended for <i>Approved as Noted</i>
				<i>Approved</i>

[PIU]

Format F-9(a)**DESIGN APPROVAL NOTE**

To
[DSC/PIU]

The enclosed designs/drawings are hereby approved.

S.No.	Details of designs/drawings

[PIU]

Format F-10**CONFORMANCE\NON-CONFORMANCE REPORT (CNC)**

Date: _____

Ref:

To
[Contractor]

Package No. : _____ Contract No.: _____

Title of works : _____

Ref: Your letter _____ dated _____.

Based on the review of the submitted test reports, as mentioned in the table below, our comments and instructions are mentioned herein for your suitable action.

Test Report No.	Date	Test Laboratory	Material	Comments <i>(Conforms/ Does not conform. See instructions below)</i>

Instructions (Actions to be taken by Contractor): _____

[PIU]

Test Reporting

Contract Package No. : _____
Contract. No. : _____

Title of Work: _____
Contractor : _____

[illegible]

Format F-12**MATERIAL REGISTER**

Contract Package No.: _____
 Contract. No. : _____
 Name of Material: _____

Title of Work : _____
 Contractor : _____
 Unit of Measure MT/Bags/Drums/M³/M²

Inward Date	Inward No.	Source of Supply	Quality	Test Report No.	CNC Report No.	Date of Issue	Quality Issued	Quality Balance	Consultant Sign	Inspecting Authority Sign
Commutative Inward Including this _____										
Commutative Inward Including this _____										

DAILY WORK RECORD/SITE ORDER BOOK

...continued page 2

DAILY WORK RECORD/SITE ORDER BOOK

[illegible]

(3 copies per set – one copy each to be sent to PIU and PMC by the contractor; one copy to be retained at site)

Format F-14**VARIATION ORDER**

Ref : VO____	Date :
Package No. :	Contract No. :
Title of Work:	
From	[PIU]
To	[Contractor]

You are hereby instructed to make the following changes in the Contract:

Description of Change

Estimated Cost of Change

Item of work	Quantity	Unit cost	Estimated cost	Change in cost

Summary of cumulative changes including this one:

- Original contract price Rs. _____
- Net of cumulative changes prior to this change Rs. _____
- Net change in this order Rs. _____

Final contract price net of all changes so far

Rs. _____

Time for completion :	Original _____ months
	Prior to this change _____ months
	Change in this order _____ months
	Final time for completion _____ months

Signatures

[PIU]

[CONSULTANT]

INSTRUCTION LOC

Title of Work :

Contractor :

[illegible]

Format F-16**MINUTES OF PROGRESS REVIEW MEETING**

Meeting Conducted on: _____

Contract No.	:	_____
Name of the Works	:	_____
Contract Amount	:	_____
Name of Contractor	:	_____
Notice to Proceed	:	_____
Contract Duration	:	_____
Completion Date	:	_____

Elapsed Time: _____ mos. _____ %

Scheduled Work Completion: _____ %

Actual Work Completed: _____ %

Compliance with commitments made during last review meeting

SL No.	COMMITMENTS BY CONTRACTOR/PIU DURING LAST REVIEW MEETING	WHETHER COMPLIED	IF NOT, WHY & WHEN WILL BE COMPLIED

Meeting Conducted on:_____

Contract No. : _____

Name of the Works : _____

Review of Progress, Quality and Coordination during this Period

Sl. No.	PROBLEMS, ISSUES, ACTIONS TO BE TAKEN	ACTION BY	DUE DATE

Meeting conducted on: _____

Contract No.	:	_____
Name of the Works	:	_____

Any Other Business / General Comments:

[Affix dated signatures]

[PIU]

[PMC]

[Contractor]

Format F-17**INTERIM EVALUATION OF CONTRACTOR PERFORMANCE**

Evaluation Conducted on : _____ **Previous Evaluation:** _____
Contract No. : _____
Name of the Works : _____
Contract Amount : _____
Name of Contractor : _____
Notice to Proceed : _____ **Work Started on:** _____
Scheduled Completion Date : _____
Elapsed Time: _____ mos. _____ % **Work Completed:** _____ %

Factors to be considered in evaluation:**Contractor's Performance this Period**

	<u>Good</u>	<u>Satisfactory</u>	<u>Poor</u>
Progress of the Works this Period	:	_____	_____
Compliance with Engineers instructions	:	_____	_____
Quality of Contractor's supervision	:	_____	_____
Quality of materials supplied	:	_____	_____
Storage and handling of materials	:	_____	_____
Quality of workmanship	:	_____	_____
Provision of adequate and skilled labor	:	_____	_____
Provision of adequate materials in time	:	_____	_____
Provision of adequate construction equipment	:	_____	_____
Provision of signboard	:	_____	_____
Site conditions/maintenance of site	:	_____	_____
Site safety and public convenience	:	_____	_____
Cooperation for inspection and measurement	:	_____	_____

CONTRACTOR'S PERFORMANCE THIS PERIOD:

The performance of the Contractor at the date of this review has **IMPROVED/ REMAINED THE SAME/ DETERIORATED** *[delete non-applicable sections]* since the last evaluation.

The above notwithstanding, the overall performance of the Contractor is judged to be **GOOD/SATISFACTORY/POOR** *[delete non-applicable sections]* up to this date.

It is hereby agreed that improvements in any deficient areas indicated above and/or as detailed on the attached list are to be immediately taken up by the Contractor, and that the Contractor's performance is to be brought up to an acceptable standard by _____ *[date]*. A further review of the Contractors performance will be conducted on _____ *[date]*, and actions will be taken as per contract in case the deficiencies have not been satisfactorily rectified.

[PIU]_____
[PMC]_____
[Contractor]

Format F-18**PARTICULARS OF CONTRACT**

Name of Works : _____
Contract No. : _____
Contractor : _____
R.A. Bill No./Date : _____
Period Covered : _____ to _____

QUALITY CERTIFICATE

This is to certify that we have inspected the conduct of the works in accordance with the established Quality Control procedures and that the items included in this Interim Payment Certificate satisfy the required quality of works and are acceptable with regard to the specifications and standards as proscribed under the Contract.

Signature/Date:

[Project Management Consultant]

ACCEPTANCE NOTE

This is to certify we accept the Consultant's Quality Certificate.

Signature/Date:

[PIU Engineer]

Note: This Quality Certificate and Acceptance Note shall be completed and attached to each Interim Payment Certificate before payment is made.

APPENDIX D

MONTHLY REPORT FORMATS

APPENDIX D.1

FORMAT FOR CONTRACTOR'S MONTHLY PROGRESS REPORTS

APPENDIX D

FORMAT FOR CONTRACTORS' MONTHLY PROGRESS REPORTS

GOVERNMENT OF ASSAM

Guwahati Water Supply Project (JICA Loan No.:IND-P201)

[PACKAGE TITLE]

PACKAGE REF. No. _____

CONTRACT No. _____

MONTHLY PROGRESS REPORT NO. _____

[MONTH, YEAR]

[NAME OF CONTRACTOR]

[PACKAGE TITLE - CITY] PACKAGE REF. No. _____
CONTRACT No. _____

Name of Contractor _____

MONTHLY PROGRESS REPORT OF CONTRACTOR

For [Month, Year]
(due on every 1st of the month)

TABLE OF CONTENTS

Table of Contents	Page No.
1. Contract Scope and Data	
1.1 Contract Scope	
1.2 Contract Data	
1.3 Contact Data	
2. Progress Summary	
3. Financial Status	
3.1 Interim Payment Certificate Summary	
3.2 Status of IPC Payments	
4. Progress of Works	
4.1 Mobilization	
4.2 Civil Works	
4.3 Plant and Equipment Supply	
4.4 Work Program	
5. Organization and Resources	
6. Quality Assurance/Quality Control	
7. Engineer's Instructions and Site Orders	
8. Variation Orders	
8.1 Variation Orders Issued	
8.2 Variation Orders Proposed	
9. Quarterly Cash Flow Forecast	
10. Other Activities and Events	
APPENDICES	
Appendix A.4.2.1	Progress Photographs – Civil Works
Appendix A.4.3.1	Progress Photographs – Plant and Equipment Supply
Appendix A.6.1	Quality Control Tests
Appendix A.6.2	Materials Certificates
Appendix A.7.1	Engineer's Instructions and Site Orders
Appendix A.8.1.1	Variation Orders Issued
Appendix A.8.2.1	Variation Orders Proposed

1. Contract Scope and Data**1.1 Contract Scope**

[Provide package name, reference number and location, and give a brief description of the works.]

1.2 Contract Data

Contract Number	
Contractor	
Employer	
Engineer	
Date of Letter of Acceptance	
Date of Contract Signing	
Date of Notice to Proceed	
Original Contract Period (months)	
Original Completion Date	
Approved Time Extension (months)	
Revised Completion Date	
Original Contract Amount (Rs.)	
Approved Total Value of Variation Orders (Rs.)	
Revised Contract Amount (Rs.)	

1.3 Contact Data

Contact Data	Contractor's Home Office	Contractor's Site Office	Employer	Engineer
Representative				
Street Address				
City				
State				
Post Code				
Country				
Telephone Number(s)				
Fax Number(s)				
E-mail				

2. Progress Summary

Item	Cumulative up to Previous Month	This Month	Cumulative to Date
Scheduled Progress (%)			
Actual Progress (%)			
Slippage (%)			
Contract Time Lapsed (months)			
Contract Time Lapsed (%)			
Interim Payment Certificate Amount (Rs.)			
Interim Payment Certificate Amount (% Contract)			

Notes:

1. "Progress" is the ratio, expressed as a percentage, of:

$$\frac{\text{Value of completed Permanent Works and approved Variation Orders}}{\text{Contract amount for Permanent Works and approved Variation Orders}}$$
2. The "Interim Payment Certificate (IPC) Amount" consists of the value of permanent works and approved variation orders, plus amounts for advances, price variation and other items, minus deductions for retention, repayment of mobilization advance, damages, etc. The value of statutory deductions for taxes, royalties, etc., which are deducted from the Contractor's payments and paid by the Employer on behalf of the Contractor, are not considered in determining the "IPC Amount".

3. Financial Status

3.1 Interim Payment Certificate Summary

IPC No.:	IPC Date:	Period:	[Month Year]	(All Amounts in Rs.)
Component	Contract Amount	Total Amount Certified Previously (Actual)	Amount Submitted this Period	Total Amount to Date
1. Mobilization Advance	—			
2. Permanent Works 2.1 Part A 2.2 Part B 2.3 Part C etc.				
3. Approved Variation Orders Nos. ___ to ___				
4. Extra for Changes in Cost and Legislation				
5. Other Items 5.1 (Describe) 5.2 etc.	—			
6. TOTAL VALUE OF WORKS (Total of items 1 through 5)	—			
7. Contract Deductions 7.1 Retention (@10% up to 5% of Contract Value) 7.2 Recovery of Mobilization Advance 7.3 Other Items (Describe) 7.4 etc.	— — — —			
8. TOTAL AMOUNT OF CONTRACT DEDUCTIONS	—			
9. INTERIM PAYMENT CERTIFICATE AMOUNT (Item 6 minus Item 8)				

3.2 Status of IPC Payments

IPC No.	Date Submitted	Amount Submitted (Rs.)	Date Paid	Amount Paid (Rs.)
Advance 1 2 etc.				
Total to Date				

Note:

1. “Amount Paid” is the amount paid for the Interim Payment Certificate or Advance, before the statutory deductions for taxes, royalties, etc., which are deducted from the Contractor’s payments and paid by the Employer on behalf of the Contractor.

4. Progress of Works

4.1 Mobilization

[Describe mobilization activities and status during report period and compare with planned activities.]

4.2 Civil Works

[Describe civil works completed in report period, and total completed to date; describe civil works in progress during report period and % completed to date; and compare actual and planned achievements. Refer to photographs in Appendix A.4.2.1.]

4.3 Plant and Equipment Supply

[Describe plant and equipment supply activities during report period, and total to date; and compare actual and planned achievements. Follow the format shown below and repeat for each major equipment item to be supplied. Refer to photographs in Appendix A.4.3.1.]

[Equipment Item]	
Activity	Description of Activities During Month / Status / % Complete
Design	
Purchase Order	
Manufacturing	
Factory Inspection and Testing	
Shipping	
Delivery	
On-Site Inspection	
Erection	
Testing	
Commissioning	
Trial Runs	

4.4 Work Program

The scheduled and actual progress of each category of works is shown in the Bar-Chart and S-Curves in the following page.

Work Program

[Insert the data on Scheduled Progress, in accordance with the approved Work Program, and on Actual Progress in the format shown below; draw the bar-charts for each category of work, showing scheduled and actual progress; and draw the S-curves of scheduled and actual cumulative progress.]

[illegible]

5. Organization and Resources

[Discuss contractor's organization, personnel and equipment employed on the project.]

6. Quality Assurance/Quality Control

[Discuss quality related issues, particularly failures. Refer to list of quality control tests and results, test sheets and materials certificates in Appendices A.6.1 and A.6.2.]

7. Engineer's Instructions and Site Orders

[Discuss instructions and site orders issued during the month and actions taken to comply with them. Refer to copies of instructions and site orders in Appendix A.7.1.]

8. Variation Orders

8.1 Variation Orders Issued

[Maintain a running account of all Variation Orders issued since the start of the contract. Attach copies of Variation Orders issued during the month in Appendix A.8.1.1.]

VO No.	Date Issued	Description	Additional Amount (Rs.)	Additional Time (days)	Remarks
Total					

8.2 Variation Orders Proposed

[Maintain a running account of proposed variation orders that have not yet been acted upon by the Engineer or the Employer. Attach copies of variation orders proposed during the month in Appendix A.8.2.1.]

Ref. No.	Date Submitted	Description	Additional Amount (Rs.)	Additional Time (days)	Remarks

9. Quarterly Cash Flow Forecast

Forecast Revision No.: _____

Date Prepared: _____

Calendar Year								
Quarter (Month-Month)	J-F-M	A-M-J	J-A-S	O-N-D	J-F-M	A-M-J	J-A-S	O-N-D
Forecast Amount (Lakhs Rs.)								
Actual Amount (Lakhs Rs.)								

10. Other Activities and Events

[Discuss other significant activities and events during the report month, such as force majeure circumstances, accidents, etc.]

APPENDICES (UNDER APPENDIX D.1)

Appendix A.4.2.1 Progress Photographs – Civil Works

[Insert photographs of milestones and important stages of civil works.]

Appendix A.4.3.1 Progress Photographs – Plant and Equipment Supply

[Insert photographs of milestones and important stages of equipment manufacturing and erection.]

[For the contractually required tests performed during the month complete the list below and attach copies of test results.]

[Contract No. _____]

[illegible]

Confirmed by:

(Consultant)

[For the contractually required materials certificates obtained during the report month complete list below and attach copies of certificates.]

Materials Certificates Obtained during [Month, Year]

[illegible]

Confirmed by:

(Consultant)

Appendix A.7.1 Engineer's Instructions and Site Orders

[Attach copies of Engineer's Instructions and Site Orders issued during the report month.]

Appendix A.8.1.1 Variation Orders Issued

[Attach copies of Variation Orders issued during the report month.]

Appendix A.8.2.1 Variation Orders Proposed

[Attach copies of Variation Orders first proposed during the report month.]